

BIOLOGY GRADE 11: TAXONOMY I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 1.0: Cytology and Biodiversity
Sub-Strand	Sub-Strand 1.1: Taxonomy I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Meaning and significance of taxonomy – Taxonomic units and nomenclature – Five Kingdoms of classification: Monera, Protocista, Fungi, Plantae and Animalia – General characteristics of the five kingdoms – Economic importance of Kingdoms Monera, Protocista and Fungi
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the significance of classification of living organisms in nature; b) illustrate the hierarchy of the major taxonomic units of classification; c) describe the principles of binomial nomenclature in classification of living organisms; d) classify organisms into the five taxonomic Kingdoms; e) appreciate the importance of taxonomy in biodiversity conservation.
Core Competencies	– Communication and Collaboration: The learner contributes to group decision-making while exploring the local environment/print media/non print media to observe and classify organisms. – Self-efficacy: The learner's confidence is enhanced as they design charts to show the hierarchy of major taxonomic units of classification and present them to peers.
Core Values	– Respect: The learner appreciates diverse opinions while discussing the principles of binomial nomenclature with peers. – Responsibility: The learner observes safety precautions while naming common organisms in the local environment using principles of binomial nomenclature.
Science & Engineering Practices	– Analyzing and Interpreting Data: Learners look for patterns in shared characteristics (cell type, organisation, nutrition) to group organisms. – Developing and Using Models: Learners build and revise a taxonomic hierarchy pyramid and a five-kingdom classification model. – Constructing Explanations: Learners defend classifications of unfamiliar organisms using observable evidence. – Engaging in Argument from Evidence: Groups compare their own classification systems and justify why a universal system is needed. – Obtaining, Evaluating and Communicating Information: Learners use textbooks, charts, videos and peer teaching (jigsaw) to

	learn about the kingdoms.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: The learner observes road safety as they explore the local environment to observe and classify organisms, and practises safe behaviour when handling specimens. – Environmental Conservation: The learner appreciates the need to conserve biodiversity while discussing the economic importance of organisms in Kingdoms Monera, Protocista and Fungi. – Health Education: Learners investigate beneficial and harmful microorganisms such as bacteria and fungi.
Career Connections	– Taxonomist: identifies, names and classifies organisms using the hierarchy and binomial nomenclature. – Botanist: studies plants such as maize and beans, for example at the National Museums of Kenya herbarium in Nairobi. – Zoologist: studies animals like goats, chickens and mosquitoes. – Ecologist: studies relationships among organisms in habitats like Lake Victoria and the Rift Valley. – Wildlife conservationist: protects biodiversity, as with Kenya Wildlife Service work. – Agricultural scientist: distinguishes crops, weeds and pests. – Medical laboratory scientist: identifies disease-causing bacteria and fungi. – Environmental scientist: monitors biodiversity and ecosystem health.
Focus for Lessons	This unit introduces taxonomy as the science of identifying, naming and classifying living organisms. Learners begin by sorting familiar Kenyan organisms in their own ways, discover that different groupings cause confusion, and progressively build the need for a universal system: the taxonomic hierarchy, binomial nomenclature and the five kingdoms. Learning is phenomenon-driven, using locally available specimens, printed photographs, prepared slides and group collaboration.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? KICD KEY INQUIRY QUESTIONS: 1. How are living organisms classified?
Anchoring Phenomenon	Groups are given a collection of familiar organisms or pictures from their local environment: a goat, a chicken, a maize plant, a mushroom, algae, bacteria, a bean plant and a mosquito. They are asked to organise them into meaningful groups. One group may sort by where organisms live, another by colour, size or usefulness (food, pests, disease), and another by whether they are green or can move. When groups compare, they find the same organisms were placed in different groups, and the same organism may have different common names in Kikuyu, Luo, Kiswahili or English. This is puzzling because everyone is correct by their own criteria, yet no one system lets people everywhere identify, study and communicate about organisms in the same way. It creates a need for a universal scientific system of classification.
Storyline Thread	Lesson 1: Introduction to Taxonomy and Classification. Living organisms can be grouped in different ways, so a systematic method is needed. Learners sort pictures or specimens of local organisms by their own criteria, compare groupings and discuss why different systems were produced. Taxonomy is introduced. Question: If organisms can be grouped in many ways, what makes scientific classification useful? Lesson 2: Importance of Classification. Classification helps scientists identify, study and communicate about organisms. Learners investigate cases such as edible and poisonous mushrooms, useful and harmful insects, and crops and weeds, and prepare reasons why classification matters. Question: If scientists all need to identify organisms accurately, what common levels or groups are used? Lesson 3: Taxonomic Units. Learners use organism and category cards to arrange ranks from broadest to most specific: Kingdom, Phylum/Division, Class, Order, Family, Genus and Species. Question: How are these taxonomic ranks related from the largest

	group to the smallest? Lesson 4: Hierarchy of Classification. Taxonomic groups form a hierarchy from broad to specific. Groups build a taxonomic pyramid or staircase and classify a familiar organism through the hierarchy using a chart or reference material. Question: How can scientists give one particular organism a unique name that is recognised worldwide? Lesson 5: Scientific Names and Binomial Nomenclature. Scientific names provide a universal way of identifying organisms. Learners compare common names in different languages and communities, discuss the problems caused, and are introduced to the two-part naming system (genus and species) and its rules. Question: How can we apply these naming rules to organisms found in our environment? Lesson 6: Applying Binomial Nomenclature. Learners apply the rules to name common local organisms correctly, writing genus and species properly, while observing safety and road safety when exploring the school environment. Question: Can characteristics shared by organisms help us group them into larger categories? Lesson 7: Five Kingdoms of Living Organisms. Organisms can be grouped into five kingdoms using shared characteristics. Using pictures, charts, specimens or videos, groups investigate one kingdom each (cell type, cell organisation, mode of nutrition, examples) and jigsaw groups teach one another. Question: Can we use the characteristics we have discovered to classify unfamiliar organisms into the correct kingdom? Lesson 8: Classification into the Five Kingdoms and Importance of Taxonomy. Classification, naming and taxonomy work together to organise biodiversity. Learners classify unfamiliar organism profiles using observable evidence, defend their classifications, revisit the anchoring phenomenon and revise their original classification. Final answer to the driving question: How can living organisms be systematically organised, named and identified?
Supporting Phenomena	– Mushrooms: some in Kenyan markets and forests are edible and others poisonous, so wrong identification can be dangerous. – The same plant or fish has different local names across communities (for example, around Lake Victoria and on the Coast), causing confusion in trade and research. – Mushrooms, algae and green-looking pond scum are often assumed to be plants, yet they have very different features. – Mosquitoes, bacteria and fungi in the community can be harmful, while others help us (yeast in mandazi dough, bacteria in mursik and yoghurt), showing why correct identification matters.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute periods)): Introduction to Taxonomy and Classification: Why Do Our Groupings Differ?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners launch the anchoring phenomenon by sorting familiar Kenyan organisms (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito) using their own criteria. They discover that every group is correct by its own rules, yet the groupings and the local names (Kikuyu, Luo, Kiswahili, English) differ, so no single system lets everyone identify and study organisms in the same way. This creates the need for taxonomy.
Knowledge	Specific Learning Outcome (a): explain the significance of classification of living organisms in nature. Learners will be able to: (1) state that classification means grouping organisms using chosen criteria; (2) explain that different criteria give different groupings; (3) define taxonomy as the science of naming, describing and classifying organisms; (4) give one reason why a universal system is needed.
Skills	NGSS Science and Engineering Practices: Asking Questions and Defining Problems (posting questions on the DQB); Analyzing and Interpreting Data (comparing group classifications); Engaging in Argument from Evidence (Claim-Evidence-Reasoning defence of criteria); Developing and Using Models (initial classification model). Core competency: Communication and Collaboration - the learner contributes to group decision-making while exploring the local environment/print media/non print media to observe and classify organisms.
Attitudes	Value - Respect: the learner appreciates diverse opinions while discussing classification criteria with peers. Value - Responsibility: the learner handles specimens and materials carefully and follows safety rules. PCI - Environmental Conservation: the learner begins to appreciate that naming and grouping organisms supports conserving biodiversity.
Key Inquiry Question	How are living organisms classified?
Purpose in Storyline	Lesson 1 opens the unit. It creates the puzzle (many valid groupings, many local names) and the need for a shared scientific system. It leads to Lesson 2: If organisms can be grouped in many ways, what makes scientific classification useful?
Safety Notes	Use printed photographs for bacteria, mosquito and algae; do not bring or culture unknown bacteria or mould. Handle any real specimens (maize, bean plant, mushroom bought from the market) only with clean hands and wash hands afterwards. Do not taste any specimen. Do not touch wild mushrooms as some are poisonous. Keep walkways clear in the classroom; if learners go outside to look at organisms, observe road safety and stay in groups within the school compound.

B. LESSON OVERVIEW

Anchoring phenomenon: Groups receive cards or specimens of eight familiar organisms: a goat, a chicken, a maize plant, a mushroom, algae (from Lake Victoria or a pond), bacteria (under a microscope picture), a bean plant and a mosquito. They sort them by their own rules (habitat, colour, size, usefulness, green or mobile). On comparing, they find the same organism in different groups and different local names (e.g. maize is 'mahindi' in Kiswahili, 'ngano ya Kikuyu' contexts vary, and names differ across communities). Everyone is right by their own criteria, yet no system lets people everywhere communicate the same way. Learners predict, sort, compare, build explanations with Claim-Evidence-Reasoning, open the Driving Question Board, and draw an initial model. Driving question: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Biodiversity and natural selection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES for similar readings	Learners sit in groups of 4-5. Each learner gets the question on the board: 'Imagine a farmer in Kericho, a fisher at Lake Victoria and a student in Nairobi each sort these eight organisms: goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito. Will they make the same groups?' Individually they write a prediction in their notebooks for 2 minutes: (1) how I would group them and why; (2) whether others will group them the same way. Then Think-Pair-Share with a partner. Several learners share predictions such as 'by where they live' or 'by what is useful for food'. They record on the board a quick tally: same groups / different groups.	Display pictures of the eight organisms on manila paper or the chalkboard. Say: 'Look at these eight living things you see around Kenya. Before we touch anything, I want your first thinking.' Read the prediction question aloud. Give WAIT TIME of 10-15 seconds of silent thinking before anyone writes or speaks. Say: 'Write one sentence. There are no wrong predictions today.' After pairs talk (2 minutes), cold-call 4 learners: 'Wanjiru, what did your partner predict?' Ask one follow-up: 'What made you think so?' Wait 10 seconds after each answer before calling on another learner. Tally the class vote on the board. Say: 'Keep your prediction. We will test it in a moment.' Circulate and listen for criteria learners use (habitat, food, colour, movement) and note unusual ones.	Predict-Observe-Explain combined with Think-Pair-Share: Writing individually before talking brings out each learner's own ideas and makes them accountable. Pair sharing lets learners rehearse and refine their thinking. The prediction becomes evidence to compare with later results, so the later surprise (many valid groupings) is something they have to explain rather than just be told.	Oral questioning and observation of notebooks. Look for: (1) a written prediction with at least one reason; (2) use of an everyday criterion such as habitat, colour, size, use or movement; (3) willingness to say whether others will agree. Record on a checklist the number of learners who predict 'same groups' versus 'different groups' and which criteria are named. Learners who leave the notebook blank after 2 minutes get a prompt: 'Which two organisms would you put together first?'
Observe Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES for similar readings	Card sorting and group investigation (about 25 minutes). Each group gets a set of 8 picture cards or specimens (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito) and manila paper. Step 1: Group agrees one rule (criterion) and sorts the cards into groups on the manila paper, writing the rule and group names. Step 2: They record what the rule is and any organism that was hard to place (e.g. mushroom, algae, bacteria). Step 3: Gallery Walk: groups rotate, one 'host' stays to explain while others visit and record on a table: Group Criterion Groups formed Where did the same organism go? Step 4: Learners note local names. Using Kikuyu, Luo, Kiswahili and English	Before distribution say: 'Your job is to make groups that make sense to your group. Choose ONE rule, write it at the top, then sort.' Remind safety: 'Specimens are handled with clean hands, nobody tastes anything, and mushrooms are not touched with bare hands.' Give 8 minutes for sorting and WAIT TIME of 10 seconds when groups say they are finished: 'Check: could someone else read your rule and sort the same way?' Circulate and ask each group: 'Why did you place the mushroom here?' and 'Which organism was hardest to place?' Do not correct the criteria. Start the Gallery Walk: 'Visit three other groups. Find one organism that was placed in a different group from yours.' Cold-	Card sorting, Group investigation and Gallery Walk: Handling the cards forces learners to decide what features matter. Comparing with other groups produces the key data: many different valid classification systems and different local names for the same organism. Seeing disagreement first-hand motivates the need for an agreed, shared system.	Observation of group work and classification card activity. Observable indicators: (1) each group has a written criterion; (2) every card is placed in exactly one group; (3) learners explain their reason using a feature of the organism; (4) each learner contributes at least one idea (check with a tally per group); (5) learners identify at least one organism placed differently by another group and list at least two local names. Use a peer check: groups tick 'rule clear/rule not clear' for the group they visit.

	names for the same organism (e.g. maize = mahindi in Kiswahili), they list names in a table. Optional 5-minute video or textbook page on 'Classification' if electricity is available; otherwise use the shared textbook and charts.	call 3 groups after the walk: 'What did you notice?' Prompt for names: 'Who can say the name for maize or chicken in your mother tongue? In Kikuyu? Luo? Kiswahili?' Write names on the board in a table, showing that names differ. Use bilingual vocabulary support for words such as 'criterion/kigezo' and 'classify/kuainisha'. Ask: 'Who is correct?' and wait 10-15 seconds.		
Explain Phase  VIDEO: Biodiversity and natural selection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES for similar readings	Learners use Claim-Evidence-Reasoning (CER) to explain the puzzle, then take part in a short discussion and debate. Individually, then in groups, they complete a CER frame: Claim: 'No single group's classification can be used by everyone.' Evidence: 'Group A put the mushroom with plants because ..., Group C put it with ... ; the name for maize is ... in ... and ... in ...' Reasoning: 'Because groups used different criteria and different names, people cannot be sure they are talking about the same organism, so we need ...' Groups then debate the question: 'Is the group that sorted by usefulness (food, pests, disease) wrong?' and 'Should the whole class vote on one system?' Teacher introduces the term taxonomy and the idea of scientific classification through a short mini-talk using a chart or video, and learners write a definition in their own words. Jigsaw/peer teaching: each member of a group takes one question (What is classification? Why classify? What is taxonomy? Why are common names not enough?) from the textbook, then returns to teach the group.	Say: 'Everyone's classification made sense. So what is the problem?' WAIT 15 seconds. Model one CER sentence for the class using the mushroom: 'My claim is ...' Then give 6 minutes for groups to write CER. Circulate and ask: 'What exact evidence from the cards or names supports that?' and 'Does your reasoning explain why?' Run the debate: assign two sides, 'one system for everyone' versus 'each community uses its own'. Say: 'Disagree respectfully: say "I see it differently because..."' Cold-call 5 learners for reasoning, using a different group each time. Then introduce: 'Scientists faced this same problem. The science of naming, describing and grouping organisms is called taxonomy.' Write it on the board with the Kiswahili support 'taksonomia/uainishaji'. Give the textbook pages to groups for the jigsaw. Correct misconceptions such as 'there is one right way for everyone'. Close with: 'What would a good system for everyone need?' Collect 3 ideas after 10 seconds of WAIT TIME.	Claim-Evidence-Reasoning, Discussion and debate, Jigsaw and Peer teaching: CER ties the claim to actual observations from the sorting activity, so learners explain from evidence instead of opinion. Debate surfaces the tension between local and universal systems. Jigsaw and peer teaching make every learner responsible for explaining one idea, which builds the vocabulary of taxonomy and gives a first answer to 'How are living organisms classified?'	Worksheet (the CER frame) and oral questioning. Indicators of success: (1) the claim states that groupings differ or that no single system works for everyone; (2) evidence cites a specific organism, group or local name; (3) reasoning links different criteria or names to difficulty communicating; (4) the learner uses the word 'criteria' or 'classification' correctly; (5) the definition of taxonomy includes naming and grouping organisms. Use peer assessment: partners swap CER and tick each of the three parts. Short quiz (2 questions) for exit: 'What is taxonomy?' and 'Give one reason common names are not enough.'
Driving	Learners open the class Driving	Say: 'Now that you have seen the	Driving Question Board: Learners'	Observation and DQB questions.

Question Board (DQB) Creation  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES for similar readings	Question Board. The driving question is written at the top on manila paper: 'How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?' Each learner writes 2 questions on sticky notes or paper strips from their observations, for example: 'What features should count?', 'Why are mushrooms not plants?', 'Are algae plants?', 'Why can bacteria be both useful and harmful?', 'Who decides the scientific name?' In groups they sort questions into categories (How to group, Names, Kingdoms, Usefulness/Harm, Conservation) and post them on the DQB. The class chooses two questions to investigate first and links them to upcoming lessons. They also post the exit question: 'If organisms can be grouped in many ways, what makes scientific classification useful?'	puzzle, what are you still wondering? Write two questions on strips, one per strip. Questions beginning with how or why are especially useful.' Give 3 minutes of silent writing with WAIT TIME of 10 seconds before allowing talk. Cold-call 6 learners to read a question and place it on the board, asking 'Which group of questions does this belong to?' Group similar questions and label categories. Highlight questions that came from learners' predictions: 'Aisha asked why mushrooms are not plants; we will test that.' Point out which questions will be answered in the next lessons and which need investigation. Leave the board visible, and say: 'We will return to this board at the end of each lesson to move or answer questions.' Link to the Lesson 2 question and ask 3 learners to restate it in their own words.	own questions become the map for the unit. Sorting questions into categories shows patterns and helps learners see what they need to learn, which keeps later lessons connected to the anchoring phenomenon and to their own curiosity.	Indicators: (1) every learner posts at least two questions; (2) questions link to the phenomenon (criteria, names, kingdoms, usefulness); (3) at least some questions are investigable, such as 'What features separate fungi from plants?'; (4) groups can justify the category they chose. Record the types of questions to plan later lessons, and give an exit ticket: 'One question I most want answered is ...'
Model Building Phase  VIDEO: Biodiversity and natural selection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES	Learners draw an initial model on manila paper or in their notebooks showing: (1) the eight organisms; (2) at least two different ways of grouping them (their group's criteria and one other); (3) a note on local names (e.g. Kikuyu, Luo, Kiswahili, English) for one organism; (4) a labelled 'problem' arrow showing what makes communication difficult; and (5) a 'what we need' box describing what a shared system should do (e.g. same names, same rules, usable by everyone). Pairs explain their models, then add a first-draft statement: 'Taxonomy is ... and it is useful because ...'. They fill the first row of the class summary table (What we observed / What	Say: 'Draw what we figured out today. Show two ways of grouping the same organisms, the problem it causes, and what a better system would need.' Give 8 minutes of drawing and WAIT TIME of 10 seconds after instructions. Circulate, asking 'Where does your model show why a farmer in Kericho and a student in Nairobi might disagree?' and 'Which part still feels unclear?' Cold-call 3 learners to share models. Emphasise that the model is a first draft and will be revised after learning about kingdoms and naming. Return to the final prompt: 'If organisms can be grouped in many ways, what makes scientific classification useful?' and ask	Model revision (first draft): Drawing a model forces learners to organise what they observed into a cause-and-effect picture of the puzzle (different criteria and names lead to confusion, so a shared system is needed). The draft provides a baseline for revision as they learn kingdoms, hierarchy and binomial nomenclature.	Exit tickets, model review and peer assessment. Look for: (1) at least two different groupings shown; (2) the problem of different names or criteria labelled; (3) a statement of what a universal system needs; (4) the definition of taxonomy used appropriately; (5) a question on the DQB linked to the model. Use a 3-point rubric: 1 = lists organisms only; 2 = shows groupings; 3 = shows groupings, the problem and the need for a universal system. Partners give one 'star' and one 'wish' comment.

for similar readings	we learned / How we explain it) and keep the model for revision in later lessons.	learners to record their best idea in the exit ticket. Collect the tickets and models to plan Lesson 2. Remind learners to use respectful language and to bring one local plant or animal name from home in their mother tongue.		
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every learner contribute a criterion and defend it with evidence? (2) Which groupings surprised me, and how can I use them in Lesson 2? (3) Did learners treat different local names with respect, and did I include Kikuyu, Luo and other languages in the room? (4) Which learners still think there is only one correct way to group organisms? (5) Which DQB questions (mushrooms, algae, bacteria) should drive the next lessons? (6) Did the practical constraints work (printed photographs, safe specimens, shared textbooks)? What would I change about timing or materials?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We sorted eight familiar organisms (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito) using our own rules. Different groups made different groupings and placed the same organism in different groups. The same organism also had different names in Kikuyu, Luo, Kiswahili and English.
What did I learn?	Organisms can be grouped in many ways depending on the criteria chosen. Taxonomy is the science of naming, describing and classifying organisms. Local names and personal criteria are useful but cannot be used by everyone in the same way.
How does this explain the phenomenon?	Each group was correct by its own criteria, but because criteria and names differ, people cannot be sure they are discussing the same organism. Scientists therefore need a universal, agreed system of classification so that everyone can identify, study and communicate about organisms in the same way.

LESSON 2 (80 minutes (2 x 40-minute periods)): Importance of Classification: Why Do We Need to Identify Organisms Correctly?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use real-life Kenyan cases (edible versus poisonous mushrooms, useful versus harmful insects, crops versus weeds) to gather evidence that classification helps scientists identify, study and communicate about organisms, and to see why a shared system is needed.
Knowledge	By the end of the lesson the learner should be able to explain the significance of classification of living organisms in nature (KICD SLO a). Learners will know that classification (1) enables accurate identification, (2) organises the huge diversity of life so it can be studied, (3) allows clear communication across languages and regions, (4) shows relationships between organisms, and (5) supports biodiversity conservation.
Skills	NGSS SEP: Engaging in argument from evidence (Claim-Evidence-Reasoning); Analysing and interpreting data (case evidence); Obtaining, evaluating and communicating information (video and printed photographs). Core competency: Communication and Collaboration, as the learner contributes to group decision-making while exploring information to observe and classify organisms.
Attitudes	Respect: the learner appreciates diverse opinions (including local names and indigenous knowledge in Kikuyu, Luo, Kiswahili and other languages) while discussing. Responsibility: the learner follows safety precautions with specimens and photographs. PCI Environmental Conservation: the learner appreciates the need to conserve biodiversity.
Key Inquiry Question	How are living organisms classified? (Lesson focus: Why do we need to classify them, and what do scientists need so they can identify organisms accurately?)
Purpose in Storyline	In Lesson 1, groups sorted goat, chicken, maize, mushroom, algae, bacteria, bean plant and mosquito in different ways, and everyone was correct by their own criteria. Lesson 2 shows that this is more than a puzzle: wrong identification can cost health, food and money. Learners build the case that classification is important. This raises the question for Lesson 3: If scientists all need to identify organisms accurately, what common levels or groups are used?
Safety Notes	Do NOT touch, taste or smell wild mushrooms; use printed photographs or video only. Use only safe, locally available specimens (maize, bean and sukuma wiki leaves, dried or preserved insects in sealed containers) and wash hands afterwards. Do not culture unknown bacteria or mould. If any outdoor observation is done, follow road safety: walk in pairs and stay on the school compound. Share textbooks in groups.

B. LESSON OVERVIEW

Learners start by predicting what could go wrong if people cannot identify organisms correctly, then investigate three case sets using a video (or printed photographs if electricity or internet is unreliable): edible and poisonous mushrooms, useful and harmful insects (honey bee, ladybird, mosquito, locust), and crops and weeds (maize versus couch grass, beans versus black jack). A jigsaw lets expert groups become peer teachers. Learners write Claim-Evidence-Reasoning statements on why classification matters, then add questions to the Driving Question Board. They revise the class model by adding a first purpose of classification. The lesson ends with the new question: if all scientists need to identify organisms accurately, what common levels or groups do they use?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Recap (5 min): Learners recall the Lesson 1 sorting of goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito, and that the same organism had different names in Kikuyu, Luo, Kiswahili and English. Teacher shows a printed photo of two similar-looking mushrooms (one edible like the oyster mushroom, one poisonous) and a farm scene with maize and weeds. Learners individually write a prediction in their notebooks: 'What could go wrong if people in Kisumu, Nairobi and Kericho identify these organisms differently?' Then Think-Pair-Share: they discuss with a partner and share with the class. Learners also predict which problems would affect health, food, money and disease control.	Say: 'Last lesson everyone was correct by their own criteria. Today we ask: does it matter if we are not all using the same system?' Display the photos and ask: 'A family in Kakamega collects wild mushrooms for supper. They call both kinds by the same local name. What might happen?' WAIT TIME 10-15 seconds in silence before any talk. Give 2 minutes of individual writing, 3 minutes of pair talk, then cold-call 4 pairs (not volunteers). Record predictions on the board under 'Our predictions' without correcting them. Prompt: 'Which prediction could we test with evidence today?' Support vocabulary by writing 'identify, classify, poisonous, edible, weed' with Kiswahili and mother-tongue equivalents beside them.	Predict-Observe-Explain and Think-Pair-Share: Writing a prediction before seeing evidence reveals learners' initial ideas (for example, 'names are enough'), creates a reason to look carefully at the cases, and lets learners compare their later explanations with their earlier thinking. Pair talk gives every learner a voice and lowers the risk of speaking in front of the class.	Oral questioning and observation. Look for: (1) predictions that name a specific consequence, such as food poisoning, spraying the wrong crop or missing a disease vector; (2) learners who mention the naming problem from Lesson 1; (3) misconceptions to note, such as 'all mushrooms are safe' or 'all insects are pests'. Tally on a checklist how many of the 4 cold-called pairs give a reasoned prediction (target: 3 of 4).
Observe Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Plant Classification (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES	Jigsaw group investigation (about 30 min). Learners form home groups of 6, then split into three expert pairs, each studying one case using the video or printed photo sets with a recording worksheet: Case A: Edible versus poisonous mushrooms (oyster and button mushrooms against deadly look-alikes; features such as cap, gills, ring, base). Case B: Useful versus harmful insects (honey bee and ladybird against mosquito, locust and tsetse fly; roles in pollination, honey, pest control, malaria and sleeping sickness). Case C: Crops versus weeds (maize and beans against couch grass and black jack/Bidens pilosa; competition for light, water and nutrients; herbicide choice). Each expert pair completes a	Introduce: 'You are the investigating team for the Ministry of Agriculture and the Ministry of Health. Your job is to find out why correct identification matters.' Distribute worksheets and photo cards, and set a 3-minute timer for the first video viewing. Circulate and ask each expert pair: 'What feature did you use to tell them apart?' and 'Who is harmed if this is wrong?' Pause the video at the key moment and WAIT TIME 10-15 seconds before taking answers. Cold-call 3 pairs for their observations. Remind learners of safety: 'Photographs only for wild mushrooms. Do not touch or taste.' Support learners with sentence starters: 'If we mistake ____ for ____, then ____.' Offer Kiswahili or mother-tongue terms alongside	Jigsaw with group investigation: Each learner becomes the expert on one case, so they must understand it well enough to teach others. This builds a bigger pattern, that identifying organisms accurately is needed across health, farming and ecology, which no single case would show. Using a video or photo evidence mirrors how scientists gather information safely.	Observation of group work and worksheets. Indicators: (1) the worksheet lists at least two identifying features per case; (2) learners state at least one consequence of wrong identification; (3) all members of the pair contribute; (4) learners use scientific words accurately (poisonous, pest, weed, vector). Use a 3-point checklist (not yet, partly, fully). Target: 80% of pairs complete the table fully.

for similar readings	table: What must be identified? What happens if identified wrongly? What features help us identify? Who needs to share this information (farmers, doctors, extension officers, researchers)? If a video is used, learners watch it twice: once for overview and once to record evidence. Learners without electricity use the photo cards and a short textbook extract shared in groups. Where safe, learners examine real maize, bean and sukuma wiki leaves and a preserved insect in a sealed container using a hand lens, noting distinguishing features.	English and invite learners to share local names for each organism, then note them on a chart to revisit later.		
Explain Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Plant Classification Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Peer teaching and Claim-Evidence-Reasoning (about 20 min). Experts return to home groups and teach their case in 2 minutes each, while others record notes in a 3-column summary table (Case, Why identification matters, Evidence). Home groups then write a Claim-Evidence-Reasoning (CER) statement answering: 'Why is classification important?' Claim: a clear statement. Evidence: data from at least two cases. Reasoning: link the evidence to the idea that grouping organisms by shared features lets everyone identify, study and communicate about them. Groups prepare reasons on a chart, for example: (1) identify organisms accurately and safely, (2) study the large number of organisms in an organised way, (3) communicate across languages (the same organism has many local names), (4) show relationships between organisms, (5) support biodiversity conservation by knowing what species exist. Gallery Walk: charts	Say: 'Teach your case in two minutes, then ask your group one question to check understanding.' Model a CER example with a non-lesson case (for example, 'Claim: farmers need to identify the fall armyworm correctly. Evidence: it damages maize but other caterpillars do not. Reasoning: correct identification leads to correct control'). Display the CER frame. Circulate and ask: 'Which piece of evidence supports your claim?' and 'How does that link to the mushroom or mosquito example?' During the Gallery Walk, set a 6-minute timer, with 2 minutes per chart. Afterwards ask: 'Which reason was strongest and why?' WAIT TIME 10-15 seconds. Cold-call 5 learners from different groups. In the debate, ensure respect: 'Listen to the person speaking, then build on or challenge the idea, not the person.' Bring back the Lesson 1 puzzle: 'Everyone was correct by their own criteria, so what is still missing?' (Answer: a common system everyone agrees on.)	Claim-Evidence-Reasoning, peer teaching and Gallery Walk: CER makes learners connect observations to the big idea instead of just listing facts. Peer teaching deepens understanding, since explaining a case reveals gaps. The Gallery Walk exposes learners to alternative reasoning and encourages respectful critique. The debate shows that local names and knowledge have value but are not enough for universal communication, which sets up the need for shared groups.	Group presentations, peer assessment and worksheets. Indicators: (1) the claim states clearly that classification helps identify, study and communicate; (2) evidence cites specific details from at least two cases (for example, the poisonous mushroom look-alike, the mosquito as a malaria vector, black jack competing with beans); (3) reasoning explicitly links the evidence to classification; (4) peer comments are specific and respectful. Use a 4-point rubric (Claim, Evidence, Reasoning, Communication). Target: most groups score 3 or more on at least 3 rows. Re-teach a CER example if groups list only facts.

	are posted, and learners tour with sticky notes giving one 'star' (strong evidence) and one 'question' (peer assessment). Discussion and debate: 'Is local knowledge enough, or do we also need a universal system?'			
Driving Question Board (DQB) Creation  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board with the driving question displayed: 'How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?' Individually they write on sticky notes (1) one thing they now understand about why classification matters and (2) one new question. Examples: 'If we all need to identify organisms accurately, what groups do scientists use?', 'How many groups are there?', 'Is a mushroom a plant?', 'Who gave organisms scientific names?', 'Why does the mosquito have a two-part Latin name?' In pairs they sort questions into clusters (for example, Groups and levels, Naming, Kingdoms). The class chooses one question to carry into Lesson 3.	Say: 'Today we learned why classification is needed. What do we still not know? Write a question that would help us build a shared system.' Give 3 minutes of silent writing, WAIT TIME 10-15 seconds before sharing. Cold-call 4 learners to read and place their question and explain where it belongs. Highlight the next-lesson question: 'If scientists all need to identify organisms accurately, what common levels or groups are used?' Circle it and add 'Lesson 3'. Ask: 'Which of our earlier questions have we partly answered?' and mark them with a tick. Honour local-name questions by posting them under a heading 'Names across communities'. Keep the board visible for future lessons.	Driving Question Board: Learners see their questions drive the learning and track what they have figured out versus what is still unknown. It links the case evidence to the anchoring puzzle of different groupings and names, and helps learners see a logical sequence toward levels of classification, kingdoms and binomial names.	Exit ticket and observation. Each learner writes a question on a sticky note that doubles as an exit ticket. Indicators: the question is testable or linked to classification (levels, groups, naming); at least 70% of questions go beyond yes/no; learners can say where a question belongs. Note learners whose questions show confusion (for example, 'Why do mushrooms have green leaves?') for follow-up.
Model Building Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES	Learners revise the class model, a large wall chart or flow diagram titled 'Why and how we organise living things'. Lesson 1 version: organisms sorted by different criteria (habitat, colour, usefulness, ability to move), with different local names. In Lesson 2 they add a new section: 'Purposes of classification' with arrows showing identify accurately, study organised groups, communicate across languages, show	Say: 'Our model must now explain why a shared system is needed. What should we add?' Give 2 minutes of individual thinking, WAIT TIME 10-15 seconds, then a pair talk. Cold-call 4 learners to propose additions, and ask the class to agree or challenge using evidence from the cases. Draw the new section on the board while learners copy. Ask: 'What can our model not yet explain?' and guide them to the gap: the groups and	Model revision: Learners make their thinking visible by adding the purposes of classification to an evolving class model. Marking the gap in the model gives a clear reason for the next lesson, helping learners see that knowledge is built step by step toward a universal scientific classification system.	Short quiz, model check and Summary Table review. Indicators: (1) the revised model includes at least 3 purposes of classification with a Kenyan example for each; (2) learners can explain why local names alone are not enough; (3) the 'missing' box correctly states the need for common groups or levels; (4) at least 75% answer 3 of 4 quiz questions correctly. Follow up with learners who list only one purpose, using the case

for similar videos  HTML: Plant Classification (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	relationships and conserve biodiversity, each tied to one Kenyan example (oyster mushroom versus poisonous look-alike, honey bee versus mosquito, maize versus couch grass). Individually they update a notebook copy and add a 'What is still missing?' box (a common set of levels and groups). Learners complete the Summary Table: What we observed, What we learned, How we can explain it.	levels everyone should use. Close with a short 4-question quiz (oral or written): (1) Name two reasons classification is important. (2) Why can local names cause confusion? (3) Give one example of a harmful insect and why identifying it matters. (4) What do we still need? Collect notebooks or the exit tickets to check understanding.		worksheets.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners use evidence from at least two cases when explaining why classification matters, or did they list facts without linking them? (2) Did the predictions show a misconception, such as 'names are enough' or 'all insects are pests', and did the evidence change their thinking? (3) Was the video or photo set accessible, and were all learners able to take part despite limited electricity or vocabulary? (4) Did the jigsaw give every learner a voice, or did some learners dominate? (5) Were local names and indigenous knowledge respected while learners recognised the need for a universal system? (6) Did the DQB questions lead naturally to levels and groups for Lesson 3? Adjust grouping, add more print resources and revisit any unclear CER statements at the start of Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe in the cases of edible and poisonous mushrooms, useful and harmful insects, and crops and weeds? Which features helped us identify each organism, and what happened when identification was wrong?
What did I learn?	What did we learn about why classification is important for identifying, studying and communicating about organisms, and for conserving biodiversity? Why are local names alone not enough?
How does this explain the phenomenon?	How can we explain, using Claim-Evidence-Reasoning, why scientists need a shared system of classification? What new question does this raise about common levels or groups?

LESSON 3 (40 minutes (one lesson; extend to a double lesson if card-making is needed)): Taxonomic Units: From Kingdom to Species

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use organism cards and category cards to arrange the taxonomic ranks from broadest to most specific (Kingdom, Phylum/Division, Class, Order, Family, Genus, Species). They see that a nested hierarchy gives every Kenyan organism one shared place, whatever its local name.
Knowledge	b) illustrate the hierarchy of the major taxonomic units of classification. Learners state the seven ranks in order and explain that each rank nests inside the one above it. Fewer organisms and more shared features occur as you move from Kingdom to Species. Kingdom and Phylum/Division are the broadest groups, and Species is the most specific.
Skills	Developing and using models (a hierarchy chart). Analysing and interpreting data (sorting cards by shared features). Constructing explanations using Claim-Evidence-Reasoning. Communication and collaboration in group decision-making.
Attitudes	Respect: learners appreciate diverse opinions while discussing how to order the ranks and the local names. Self-efficacy: learners grow in confidence as they design charts showing the hierarchy and present them to peers. Responsibility: learners handle materials and the classroom space carefully.
Key Inquiry Question	How are living organisms classified? Lesson question: How are these taxonomic ranks related from the largest group to the smallest?
Purpose in Storyline	In Lessons 1 and 2, learners found that sorting by habitat, colour or usefulness, and using local names, gives conflicting groups. In this lesson they see that a universal system needs ranked, nested levels. The lesson leads to the next question of how the ranks are related from largest to smallest, and to later work on binomial nomenclature (Genus + species) and the Five Kingdoms.
Safety Notes	Card activity is indoors, so keep walkways clear of bags. Use printed photographs and cards rather than live specimens. If learners are sent outside to find examples, they observe road safety, stay in pairs and keep away from roads, and they do not touch mushrooms, unknown plants or insects such as mosquitoes. Learners wash hands after any handling. Where electricity or internet is unreliable, use printed screenshots from the video or a teacher read-aloud summary.

B. LESSON OVERVIEW

Anchoring phenomenon: groups sorted a goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito in different ways, and the same organism had different names in Kikuyu, Luo, Kiswahili and English. Everyone was right by their own criteria, but no shared system existed. In Lesson 3 learners predict how a scientist could place any organism in nested groups. They watch the video 'Taxonomy | Classification of Living Organisms' (or use printed stills) and then work with organism cards and category cards. They arrange the seven ranks from broadest to most specific and test the order with a Kenyan example, the maize plant (*Zea mays*) beside the lion (*Panthera leo*) or the domestic cat. Learners explain the hierarchy using Claim-Evidence-Reasoning, add new questions to the Driving Question Board, and revise the class model so that it now has ranked levels. Materials per group of 4-5: one set of 7 rank cards, 2-3 organism 'ladder' card sets, a printed classification diagram, a mahjong or manila paper sheet, markers, and shared textbooks. Vocabulary support: key terms are written in English with Kiswahili support (Ufalme = Kingdom, Jamii/Familia = Family, Jenasi = Genus, Spishi = Species), and learners add local names.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy taxonomic ranks kingdom	Learners recall the sorting task with the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito. They are told that scientists place every organism in a series of groups inside groups, like a Kenyan address: Country, County, Sub-county, Ward, Village, Plot. Individually, learners write a prediction: 'If a scientist wanted to place a goat in the most exact group possible, what groups might they use and in what order, from biggest to smallest?' They then Think-Pair-Share, and each pair draws a quick sketch of their predicted order with a reason.	Display the 8 organism pictures and ask: 'Last lesson, the same mandazi-sized problem came up: one animal, many names and many groups. How could we make one system everyone uses?' Then say: 'Think about how we find a home in Kenya. Which comes first, Kenya or the village? Write your own prediction on your sheet.' Give WAIT TIME of 10-15 seconds after each question. Give 2 minutes for silent individual writing, then 2 minutes of pair talk. Cold-call 4 pairs (not volunteers): 'What order did you predict, and what is your reason?' Record all predictions on the board without judging them. Say: 'I'm not telling you yet. We will test these ideas with evidence.' Circulate and note which pairs use the address idea and which use size alone.	Predict-Observe-Explain with Think-Pair-Share. The prediction makes learners' prior ideas about size, groups and order visible and gives them a stake in the evidence. The address analogy links to familiar nested structures, so the hierarchy becomes meaningful and not just a list to memorise.	Oral questioning and observation of group work. Look for: (1) writing or sketching a prediction with a stated reason (not just 'I think so'), (2) any learner using 'biggest to smallest' or 'more general to more specific' language, (3) pairs who both speak during the share. Note misconceptions such as 'Species is the biggest group' or 'order by animal size'. Tally on a quick class list: Prediction made with reason / Prediction without reason / No prediction. Use this to decide who needs extra support during card sorting.
Observe Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy taxonomic ranks kingdom	Learners watch the video 'Taxonomy Classification of Living Organisms' (or study printed stills and a teacher read-aloud if there is no electricity) and take notes using a guided worksheet: names of the ranks, what happens to the number of organisms as you move down, and an example. Then in groups of 4-5 they do the Card sorting investigation. Each group has 7 rank cards (Kingdom, Phylum/Division, Class, Order, Family, Genus, Species) and sets of organism cards for the goat/cow, maize and lion/cat. Step 1: arrange the rank cards from broadest to most specific. Step 2: match the organism-category cards to each rank for one animal (for example Animalia, Chordata, Mammalia, Carnivora, Felidae,	Introduce the video with a purpose question: 'Watch for two things: the order of the ranks, and what happens to the number of organisms as we move down.' Pause the video twice and ask: 'What rank comes right after Kingdom?' Cold-call 3 learners after each pause with WAIT TIME of 10-15 seconds. Distribute the card packs and say: 'Place your cards from the broadest group at the top to the most specific at the bottom. Agree as a group before you move a card, and give a reason each time.' Circulate and ask: 'Why does Family come before Genus? How do you know?' and 'How many organisms could belong to this rank compared with the one below?' Do not correct the order directly; ask 'What evidence	Card sorting and Jigsaw. Card sorting turns an abstract list into a physical model that learners can rearrange and test, and exposes ordering errors quickly. Jigsaw makes each learner responsible for part of the hierarchy and gives practice in peer teaching and communication. Linking the funnel counts to the phenomenon shows why a nested system solves the problem of the goat, chicken, maize and mosquito being grouped differently by each group.	Classification card activities and observation of group work. Observable indicators: (1) the rank cards are in the correct order (Kingdom, Phylum/Division, Class, Order, Family, Genus, Species) with a reason for each move, (2) the organism cards match their ranks correctly for at least one organism, (3) each learner teaches their section accurately in the Jigsaw, (4) the funnel diagram shows decreasing numbers of organisms. Use a checklist with 3 levels: Secure (full ladder correct and explained), Developing (1-2 ranks swapped), Beginning (needs prompts). Collect the worksheet for later review.

	Panthera, Panthera leo) and one plant (Plantae, Angiospermophyta, Monocotyledonae, Poales, Poaceae, Zea, Zea mays). Step 3: count how many organisms in the class's picture set would fit in each rank and record on a 'funnel' diagram. Jigsaw: each group member becomes expert on one section: (A) Kingdom and Phylum/Division, (B) Class and Order, (C) Family, (D) Genus and Species. They teach their section to the others and reassemble the whole ladder. Learners record how the cards changed as they moved down: fewer organisms and more shared features.	from the video supports that?' When a group is ready, check the order and ask them to explain it. For Jigsaw, say: 'Move to your expert group and prepare to teach in 3 minutes; then return and teach your home group.' Remind: 'Respect every voice; everyone must be able to explain the whole ladder.' Support language learners by pairing English terms with Kiswahili and local names (e.g. mbuzi, ng'ombe, mahindi, simba). For fast groups, add a challenge card: 'Where would a bean plant (Phaseolus vulgaris) split from maize?'		
Explain Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy taxonomic ranks kingdom	Learners use Claim-Evidence-Reasoning to answer: 'How are the taxonomic ranks related from the largest group to the smallest?' Individually they write a claim (for example, 'Each rank is nested inside the one above it, and groups get smaller and more alike as we move down from Kingdom to Species'), cite evidence from the video and their cards (the lion and maize ladders, funnel counts), and give reasoning (why nesting means any organism has one unique place, so a Kikuyu, Luo or Kiswahili speaker can find the same organism). Then they do a Gallery Walk: groups post their ladder and funnel diagram, and others walk round leaving sticky-note feedback ('I agree because...' and 'I wonder about...'). Finally, a short class discussion and friendly debate: 'Should we sort organisms by where they live as a main rank? Why or why not?' Learners use their evidence to support their positions respectfully.	Display the CER frame: 'My claim is... The evidence is... This is because...' Say: 'Use at least two pieces of evidence from the cards or the video.' Give 3 minutes of silent writing. Then cold-call 3 learners to read their claims, with WAIT TIME of 10-15 seconds before each. Ask: 'How does this ladder help solve our problem of different names for the same animal?' For the Gallery Walk, set 6 minutes: 'Move clockwise every 90 seconds. Leave one comment that helps and one question.' Circulate and listen for accurate use of 'broadest', 'most specific', 'nested' and 'shared features'. Prompt debate: 'Some say habitat could be a rank. Who can use the evidence to respond?' Remind learners: 'Disagree with the idea, not the person.' Clarify misconceptions directly: Species is the most specific rank, not the largest; Phylum is used for animals and Division for plants. Provide a mnemonic if needed, such as 'King Philip Came Over For Good Soup',	Claim-Evidence-Reasoning with Gallery Walk and Discussion and debate. CER requires learners to connect the card evidence to the idea of nested ranks and to the phenomenon of conflicting local systems. The Gallery Walk exposes them to alternative representations and strengthens peer assessment. The debate helps them see why a universal hierarchy is more reliable than habitat, colour or usefulness alone.	Worksheets, taxonomic hierarchy diagrams and peer assessment. Rubric for the CER (0-3): claim states nesting and broadest-to-specific order; evidence cites at least two specific data points (for example the funnel counts or the lion ladder); reasoning links to why a shared system is needed. Peer assessment: sticky notes must include one specific strength and one question. Observable indicators in discussion: correct use of the terms 'broadest' and 'most specific', respectful turn-taking, and evidence-based responses. Record names of learners who still reverse Genus and Family or confuse Phylum with Division, for follow-up in the exit ticket.

		and invite learners to invent a Kiswahili or Kenyan version (for example 'Kila Pishi Ciarwa Omba Fanya Gawa Sukuma').		
Driving Question Board (DQB) Creation  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy taxonomic ranks kingdom	Learners return to the class Driving Question Board: 'How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?' In groups they write on sticky notes (a) one thing they can now answer, with the evidence (for example 'The seven ranks go from Kingdom to Species and each nests inside the one above'), and (b) at least one new question (for example 'How do we write the name Genus + species?', 'Why do some books say Phylum and others Division?', 'How many Kingdoms are there and which one do mushrooms, algae and bacteria belong to?'). They place notes on the DQB under 'Answered' or 'New questions'. Each learner also completes a short exit ticket: name the ranks in order and give one reason the system helps people in different parts of Kenya.	Say: 'Look back at the board. What can we now answer? What new questions has today raised?' Give 2 minutes of quiet writing, then WAIT TIME of 10-15 seconds. Cold-call 4 groups to read one note and place it, asking: 'Which part of the driving question does this help with?' Cluster similar questions visibly: ranks, naming, Kingdoms, tools to observe. Point out the question that leads to the next lesson: 'Several of you asked how the ranks are related from the largest group to the smallest and how the names are written. We will keep working on that.' Collect exit tickets at the door. Praise specific contributions to build self-efficacy: 'Wanjiku's question about Phylum and Division shows she noticed a real difference between plants and animals.'	Driving Question Board. The DQB keeps a visible record of the class's growing understanding and shows learners that their questions guide the unit. Sorting notes into 'answered' and 'new questions' helps them see progress toward the driving question and the need for the next piece of evidence.	Exit tickets and short quiz items. Success criteria: lists all 7 ranks in the correct order (allow minor spelling errors), identifies Species as the most specific and Kingdom as the broadest, and gives a relevant reason the system helps (for example the same organism can be identified by anyone regardless of local name). Sticky-note quality: includes a specific question tied to the lesson, not just 'I want to know more'. Sort exit tickets into three piles (secure, partly secure, not yet) and plan a 5-minute reteach with card sets for the last pile at the start of the next lesson.
Model Building Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES</i>	Learners revise the class model, a large chart that began as groups of sorted organisms in Lesson 1 and was extended in Lesson 2. In groups they add a new 'nested ladder' section to the model, drawing the seven ranks from Kingdom to Species and writing the lion (<i>Panthera leo</i>) and maize (<i>Zea mays</i>) examples beside it. They add arrows to show 'is inside' and a note: 'Fewer organisms and more shared features as we go down.' They label what the model now explains (why one system can place all organisms in one exact	Say: 'Our class model began with groups that did not agree. Today we have new evidence. How should we change the model?' Give each group a marker and the model section. Prompt: 'Show the order of the ranks and show how each group sits inside the one above it.' Circulate with WAIT TIME of 10-15 seconds after questions such as 'What does your model now explain that it could not before? What is still missing?' Cold-call 3 groups to present and 2 other learners to restate one improvement. Ask: 'Where would	Model revision with peer teaching. Learners update a shared model with new evidence, which shows how scientific ideas develop over time. Explaining their revisions to others consolidates understanding and exposes remaining gaps that motivate the next lesson.	Taxonomic hierarchy diagrams, group presentations and observation. Indicators: model includes all 7 ranks in the correct order, shows nesting with arrows or boxes, includes at least one correct Kenyan example, and explicitly states one limitation or unanswered question. During presentations, check that every group member can explain at least one part. Use a quick 3-2-1 self-rating from learners (3 ranks I am sure of, 2 I am unsure of, 1 question). These feed into the final classification challenge later in the

<i>library for this activity.</i>  Search ARES for readings Search terms: taxonomy taxonomic ranks kingdom	position) and what it cannot yet explain (how the Genus and species names are written, and how organisms are placed into the Five Kingdoms). Each group presents its revision in 1 minute, and the class agrees on a final shared version.	the goat, chicken, bean plant and mosquito fit on this ladder? Which ranks do they share?' End by linking forward: 'Next, we look at how the ranks are related from the largest group to the smallest, and how two-part names give each species its own label.' Check that the final model shows the correct order and that all learners can explain it.	unit.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners arrange the seven ranks with reasons, or did they memorise the order without understanding nesting? (2) Which ranks were most often swapped (Family and Genus? Class and Order?), and which card or example would clarify this? (3) Did the maize and lion examples help learners see that local names do not matter in the hierarchy? (4) Were all learners, including quieter ones and those still building English vocabulary, involved in the Jigsaw and Gallery Walk? (5) Were printed materials sufficient when electricity or internet failed? (6) Which DQB questions should shape Lesson 4 (relations among ranks, binomial nomenclature)? Note learners needing more support and adjust the groups for the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We used organism cards and category cards to arrange the ranks from broadest to most specific: Kingdom, Phylum/Division, Class, Order, Family, Genus and Species. We watched the video on classification of living organisms and saw how the lion (<i>Panthera leo</i>) and maize (<i>Zea mays</i>) fit into the ranks. As we moved down, there were fewer organisms in each group and they shared more features.
What did I learn?	The taxonomic ranks are nested: each rank sits inside the one above it. Kingdom is the broadest group and Species is the most specific. Every organism can be given one exact place in this hierarchy, no matter what name it has in Kikuyu, Luo, Kiswahili or English.
How does this explain the phenomenon?	This helps solve our puzzle because a ranked, universal hierarchy lets everyone in Kenya and the world place, identify and talk about the same organism in the same way, unlike sorting by habitat, colour or usefulness. We still need to learn how the ranks relate in more detail from the largest group to the smallest and how organisms are named with Genus and species.

LESSON 4 (80 minutes (2 x 40-minute periods)): Hierarchy of Classification: Climbing the Taxonomic Staircase

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners build a taxonomic staircase showing how groups nest from broad (Kingdom) to specific (Species) and classify a familiar Kenyan organism through every level. This helps them see why a shared, ranked system lets people everywhere identify and study organisms in the same way, and it raises the need for a unique worldwide name for each organism.
Knowledge	By the end of the lesson the learner should be able to: (b) illustrate the hierarchy of the major taxonomic units of classification (Kingdom, Phylum, Class, Order, Family, Genus, Species); explain that groups become smaller and more similar as one moves from Kingdom to Species; and place a familiar organism (e.g. goat, maize, domestic chicken) at each level using a chart or reference material.
Skills	NGSS SEP: Developing and Using Models (building a staircase/pyramid), Analyzing and Interpreting Data (reading a classification chart), Constructing Explanations (Claim-Evidence-Reasoning), Engaging in Argument from Evidence. Core competencies: Communication and Collaboration (contributing to group decisions while classifying organisms); Self-efficacy (designing charts showing the hierarchy and presenting them to peers).
Attitudes	Respect: appreciating diverse opinions when groups justify different placements and mnemonics. Responsibility: handling specimens and materials safely and returning them. PCI Environmental Conservation: appreciating that an organised system helps us know and protect Kenya's biodiversity. PCI Safety and Security: observing road safety if specimens are collected outside the classroom.
Key Inquiry Question	How are living organisms classified?
Purpose in Storyline	Lessons 1-3 showed that sorting by colour, habitat or local names (Kikuyu, Luo, Kiswahili, English) gives different groupings, so a universal system is needed. Lesson 4 gives the first piece of that system: a ranked hierarchy that everyone can use. It ends with a new puzzle: even when two people agree on the Genus and Species groups, what single name can scientists give one organism so it is recognised worldwide? This leads to Lesson 5 (binomial nomenclature).
Safety Notes	Use printed photographs or safe, locally available specimens only (maize plant, bean plant, dried mushroom picture). Do not taste or collect unknown mushrooms, and do not culture bacteria or mould. If learners collect specimens outside, follow road safety: stay in pairs, keep away from roads, and walk in a line. Wash hands after handling plants or soil. Share textbooks in groups. If internet or electricity is unreliable, use printed screenshots from the video and the printed chart.

B. LESSON OVERVIEW

Learners return to the opening puzzle: the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito were sorted in many different ways, and the same organism had different names in Kikuyu, Luo, Kiswahili and English. In this lesson they predict how a worldwide system might 'nest' groups inside bigger groups, watch a short video (Video 2) and use a chart or reference page to gather evidence, then build a physical taxonomic staircase or pyramid with seven cards (Kingdom to Species). Using Jigsaw, each expert group classifies one Kenyan organism (goat, domestic chicken, maize, bean plant, mosquito, mushroom), then teaches peers in a Gallery Walk. Learners write a Claim-Evidence-Reasoning statement about why the hierarchy helps everyone identify organisms in the same way. They add to the Driving Question Board and revise their class model, ending with the new question: How can scientists give one particular organism a unique name that is recognised worldwide?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy hierarchy kingdom phylum	Learners recall the opening puzzle (goat, chicken, maize, mushroom, algae, bacteria, bean plant, mosquito). Working individually then in pairs (Think-Pair-Share), they draw a quick sketch predicting how a worldwide system could arrange groups so a goat in Kisumu and a goat in Nairobi are placed in the same spot. They predict: Are groups all the same size? Do some groups sit inside others? They write a one-sentence prediction in their notebooks: 'I think the groups in a scientific system are arranged like _____ because _____. ' (10 minutes)	Display pictures of a goat (mbuzi), chicken (kuku), maize (mahindi) and mosquito (mbu). Say: 'Last lesson we saw that people in Kikuyu, Luo and Kiswahili communities may call the same organism by different names. Today we ask: how could groups be arranged so everyone puts the goat in the same place?' Give 1 minute of silent THINK time, then 2 minutes PAIR talk, then WAIT TIME of 10-15 seconds before cold-calling 4 pairs to SHARE. Ask: 'What shape or picture did you use: a list, a ladder, a pyramid, a set of boxes inside boxes?' Record every prediction on the board without judging. Say: 'Hold on to your prediction. We will test it with evidence.' Prompt for reasoning: 'What makes you think groups go inside bigger groups?'	Think-Pair-Share with Predict-Observe-Explain. Making a prediction first exposes learners' existing ideas (e.g. 'a list of equal groups') and creates a reason to look for evidence. Sharing in pairs lets quieter learners rehearse their ideas, and the recorded predictions can be compared with the observations later.	Oral questioning and observation of pair talk. Look for: (1) learners who draw nested, ladder or pyramid shapes versus flat lists; (2) learners who give a reason beyond 'it looks neat'; (3) misconceptions such as 'all groups are equal in size' or 'classification is by colour'. Tally on a clipboard how many pairs predicted a nested/ranked arrangement. Use this tally to decide how much support to give in the Observe phase.
Observe Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy	Part A (15 minutes): Learners watch Video 2 on the hierarchy of classification (or study printed screenshots if electricity or internet is unreliable) and complete a note-catcher: the order of the seven levels, what happens to the number of organisms and the level of similarity going down the hierarchy, and one example organism. Part B (25 minutes): In groups of 4-5 (Group investigation + Card sorting), learners receive 7 blank cards (Kingdom, Phylum, Class, Order, Family, Genus, Species) and a shared chart or textbook page. They arrange the cards into a taxonomic staircase or pyramid on a sheet of manila paper. Each group is then	Before the video say: 'Watch for three things: the order of the levels, what happens to the group size as we go down, and the example organism. Write them in your note-catcher.' Pause the video twice and ask: 'What is the biggest group? What is the smallest?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. For Part B, model the first level on the board with the goat: 'All animals sit in Kingdom Animalia, so this is the widest step.' Circulate and ask each group: 'How many kinds of organisms do you think belong in this step compared with the step below it?' and 'What feature do the organisms in this group share that	Card sorting and Group investigation. Physically placing the seven ranks in order, then fitting a real Kenyan organism into every step, makes the abstract idea of 'nested groups' concrete. The staircase or pyramid is a model: learners see that each step down has fewer organisms with more shared features. Using a common chart or reference means all groups are working from the same evidence, linking back to the puzzle of different names and criteria.	Observation of group work using a checklist: (1) cards in the correct order, Kingdom at the broadest level and Species at the narrowest; (2) every level of the assigned organism filled in using the reference, not guessed; (3) each group member contributes a role; (4) learners can point to a step and say 'this group is bigger than the one below'. Also check note-catchers for correct ordering of the seven levels and a statement of how similarity changes. Ask 'What would happen to the number of organisms if we moved from Family to Genus?' to listen for 'fewer'. Collect classification cards as evidence.

hierarchy kingdom phylum	assigned one Kenyan organism (goat, domestic chicken, maize plant, bean plant, mosquito, mushroom) and uses the chart or reference material to fill in each level. Example for the goat: Kingdom Animalia, Phylum Chordata, Class Mammalia, Order Artiodactyla, Family Bovidae, Genus Capra, Species Capra hircus. Learners draw one or two features at the top and bottom levels to show how groups become more specific.	the ones outside do not?' Use a mnemonic prompt such as 'King Philip Came Over For Good Soup' and invite groups to create Kiswahili or mother-tongue versions. Assign roles: reader, recorder, card arranger, spokesperson. Provide vocabulary support with a word wall of the 7 level names and translations. Remind learners to share textbooks and handle printed pictures carefully. If groups finish early, they classify a second organism (e.g. maize: Plantae, Angiosperma phylum/division, Monocots, Poales, Poaceae, Zea, Zea mays).		
Explain Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy hierarchy kingdom phylum	Learners use the Jigsaw strategy (25 minutes). Expert groups (the Observe groups) first agree on a 2-minute explanation of their organism's classification. New mixed groups are formed with one expert from each organism, and each expert teaches the others (Peer teaching) using their staircase. Mixed groups then compare: Which levels did the goat and the chicken share? (Kingdom Animalia, Phylum Chordata). Where did they separate? (Class: Mammalia vs Aves). They then write a Claim-Evidence-Reasoning (CER) statement: Claim - 'A ranked hierarchy helps everyone classify organisms in the same way.' Evidence - the staircase of their organism and the comparison with at least one other organism. Reasoning - why shared levels show relationship and why a single agreed order removes the confusion from different local names or sorting criteria. Finally, a short Gallery Walk: groups post	Say: 'You are now the expert. Your job is to teach your organism's staircase to others in 2 minutes without just reading the card.' While groups teach, circulate and ask: 'At which step did the goat and the chicken separate? What does that tell you about how closely related they are?' Prompt for connection to the phenomenon: 'Suppose a group in the first lesson had put the mosquito and the chicken together because both can fly. Where would a scientist place them in this system, and why does that differ from sorting by ability to fly?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 5 learners from different groups to share a CER statement. Use sentence frames on the board: 'I claim that _____. My evidence is _____. This shows _____ because _____.' During the Gallery Walk say: 'Leave one star (something correct and clear) and one question. Be respectful of different ways of presenting the same hierarchy.' Address	Jigsaw with Peer teaching, Claim-Evidence-Reasoning and Gallery Walk. Teaching someone else forces learners to organise their understanding. Comparing organisms across groups lets them see the pattern that organisms share broad levels and differ at narrower levels. CER links the evidence (the staircase) to the reasoning about why a universal ranked system solves the earlier problem of different names and criteria.	Peer assessment (star and question notes), group presentations and CER statements. Indicators of success: (1) the CER claim states that the hierarchy gives a shared, ordered system; (2) the evidence names at least two levels where two organisms are the same or different; (3) the reasoning connects to the opening puzzle of different names or criteria; (4) learners use the correct terms Kingdom to Species. Collect CER statements and grade on a 3-point scale (claim, evidence, reasoning). Use short oral questions such as 'What is the order from broadest to most specific?' with 3 cold calls to check the sequence.

	staircases and leave one sticky-note comment (one star, one question) on two other posters (Peer assessment).	misconceptions such as 'Species is the biggest group' or 'Genus and Species are the same'. Clarify that species is the most specific level and that the idea of a unique name for each organism is still unresolved.		
Driving Question Board (DQB) Creation  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy hierarchy kingdom phylum	Learners return to the class Driving Question Board (How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?). Individually they write on sticky notes (a) one thing they now understand about the hierarchy (e.g. 'Groups get smaller and more similar as we go from Kingdom to Species') and (b) one new question (e.g. 'Why does the goat have a Latin name like Capra hircus?' or 'How do we write the name so it is the same in Kisumu and in Mombasa?'). They post notes under 'Answered' or 'New questions'. The class then identifies the question that links to the next lesson. (10 minutes)	Say: 'Look at the Driving Question Board. Which of our earlier questions can we now answer or partly answer?' Give WAIT TIME of 10-15 seconds, then invite 3 learners to move a question to the 'Answered' column with evidence. Say: 'Now write one new question. The staircase ends at Species, but people in different communities still use different names. What do you wonder?' Cold-call 4 learners to read their new questions. Highlight or group questions that point to naming and ask: 'How can scientists give one particular organism a unique name that is recognised worldwide?' Write it on a large card and place it in the centre as the lead-in question for Lesson 5. Praise respectful listening and diverse questions.	Driving Question Board. The DQB makes learners' thinking visible and shows progress from the opening puzzle. Moving questions to 'answered' with evidence, and generating new questions from a gap in the model, helps learners see that science is a process of building and revising understanding and gives them ownership of the next investigation.	Exit tickets and DQB contributions. Each learner completes an exit ticket: (1) list the seven levels in order, (2) classify one named organism down to Family or Species using the chart, (3) write one remaining question. Look for: correct order, correct use of the chart, and a new question connected to naming. Sort tickets into three piles (secure, nearly there, needs support) to plan the start of Lesson 5.
Model Building Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading</i>	Learners revise the class model for the Driving Question. In groups, they update a model (poster or diagram) that began in earlier lessons (showing that sorting by colour, use or local names gives different results and that a Kingdom-level split into groups is useful). They add the new component: a taxonomic staircase from Kingdom to Species with the goat or maize as the worked example, and write a short caption: 'A hierarchy lets everyone place an organism in the same position.'	Say: 'Take out your model from the last lesson. What can it explain now that it could not before? What can it still not explain?' Give WAIT TIME of 10-15 seconds. Provide coloured pens so revisions are visible. Circulate and ask: 'Where on your model do you show that two organisms can share a Kingdom but differ at Genus?' and 'If a farmer in Kericho and a student in Kisumu both used this staircase, what problem would be solved and what problem would remain?' Cold-call 3 groups to	Model revision. Updating a visible model forces learners to integrate new evidence (the hierarchy) with earlier ideas (the problem of different criteria and names). Marking what the model still cannot explain highlights the gap that motivates Lesson 5 and shows that models are tools which improve over time.	Taxonomic hierarchy diagrams and the final model revision. Success indicators: (1) the hierarchy is added in the correct order; (2) at least one organism is shown at each level; (3) the caption explains why the system helps everyone; (4) a gap or question about unique naming is identified. Use the Final classification challenge informally: give a picture of a new organism (e.g. a tilapia from Lake Victoria) and ask groups to name the Kingdom, Phylum and Class using

<i>in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy hierarchy kingdom phylum	They mark a gap with a question mark: 'We still need one unique name for each species.' Groups then complete the summary table (what we observed, what we learned, how it explains the phenomenon). (10 minutes)	explain one revision. Close by saying: 'We can now place an organism in the same position everywhere, but the last step, the Species name, still needs a rule. Next lesson we will ask: How can scientists give one particular organism a unique name that is recognised worldwide?'	the chart. Record the results to plan support for Lesson 5.
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D. TEACHER REFLECTION

Which groups placed the seven levels in the correct order without the chart, and which needed support? Did learners understand that groups become smaller and more similar from Kingdom to Species, or did some treat all levels as equal in size? How well did learners connect the staircase to the opening puzzle of different names and criteria? Which vocabulary or bilingual supports (word wall, Kiswahili or mother-tongue mnemonics) helped most? If electricity or internet failed, did the printed screenshots give the same evidence? What will I change to give quieter learners more speaking opportunities in the Jigsaw? Which exit-ticket responses show learners ready for binomial nomenclature in Lesson 5?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that organisms can be arranged into seven ranked groups (Kingdom, Phylum, Class, Order, Family, Genus, Species). When we classified the goat, chicken, maize plant, bean plant, mosquito and mushroom, some shared the same Kingdom or Phylum but separated at lower levels.
What did I learn?	We learned that taxonomic groups form a hierarchy from broad to specific: each step down has fewer organisms that are more similar to each other. Species is the most specific level. Using a common chart or reference lets people classify organisms the same way.
How does this explain the phenomenon?	This helps explain the opening puzzle: sorting by colour, habitat or local names (Kikuyu, Luo, Kiswahili, English) gives different groups, but a shared ranked hierarchy lets everyone place an organism in the same position. We still need a way to give each species a unique name recognised worldwide: How can scientists give one particular organism a unique name that is recognised worldwide?

LESSON 5 (80 minutes (2 x 40-minute periods)): Scientific Names and Binomial Nomenclature: One Name for Every Organism

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use evidence from the common names of familiar Kenyan organisms in different languages to figure out why a universal two-part scientific name is needed, and learn the rules of binomial nomenclature so that people everywhere can identify and study organisms in the same way.
Knowledge	Learners describe the principles of binomial nomenclature: (1) a name has two parts, the genus (capital first letter) and the species (small letters); (2) names are in Latin or Latinised form; (3) names are written in italics when typed or underlined separately when handwritten; (4) each organism has only one accepted scientific name used worldwide. Examples: <i>Zea mays</i> (maize), <i>Phaseolus vulgaris</i> (bean), <i>Capra hircus</i> (goat), <i>Gallus gallus domesticus</i> (chicken), <i>Anopheles gambiae</i> (mosquito), <i>Homo sapiens</i> (human).
Skills	Analysing and interpreting data (comparing name tables), Obtaining, evaluating and communicating information (video and peer teaching), Constructing explanations using Claim-Evidence-Reasoning, Developing and using models (revising the class classification model).
Attitudes	Respect: learners appreciate diverse opinions and local-language names while discussing the principles of binomial nomenclature with peers. Responsibility: learners observe safety and care when handling any specimens or pictures. Communication and Collaboration: learners contribute to group decision-making. Self-efficacy: learners gain confidence by presenting and correctly writing scientific names to peers.
Key Inquiry Question	How are living organisms classified?
Purpose in Storyline	Lessons 1-4 showed that groups sort organisms by different criteria and that the same organism has different names in Kikuyu, Luo, Kiswahili and English, so no single system works for everyone. Lesson 5 provides the naming part of the answer: a universal two-part scientific name. It raises the new question for Lesson 6: How can we apply these naming rules to organisms found in our environment?
Safety Notes	Use printed photographs or safe, locally available specimens only (maize, bean, goat or chicken pictures). Do not handle or culture unknown bacteria, mould or mushrooms; wash hands after handling any plant material. If learners move around the school compound, remind them to stay within the school boundary and observe road safety. Where electricity or internet is unreliable, use the printed video summary cards instead of the video: https://www.youtube.com/watch?v=vRCdOOrfuv8

B. LESSON OVERVIEW

Learners return to the original phenomenon: the same organisms had different names in Kikuyu, Luo, Kiswahili and English, so groups could not communicate. In this lesson they predict what problems arise from common names, gather evidence from a name-comparison card table and a video on binomial nomenclature, use Jigsaw and peer teaching to master the rules, write Claim-Evidence-Reasoning explanations, add to the Driving Question Board, and revise the class model so that scientific names become the universal label. The lesson ends with the question: How can we apply these naming rules to organisms in our own environment?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(Approx. 12 minutes) Learners sit	Hold up the picture of the mosquito	Predict-Observe-Explain combined	Oral questioning and observation

 VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy scientific name binomial  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy scientific name binomial	in groups of 4 with a Name Card Set (printed pictures with names) showing: maize (mahindi in Kiswahili, mbembe/ngano ya Kikuyu names such as 'mbembe' vary, 'oduma' in Dholuo), the bean (maharagwe, mboco, oganda), the mosquito (mbu, njuguma/ngoro varies, 'nyinyi' varies by community) and the goat (mbuzi, nthenge/mbūri, diel). Note: teacher fills in the local names that learners and the school community actually use; learners add the names they know. Individually, learners write a prediction in their notebooks: 'If a farmer from Kisumu and a researcher from Nairobi each use only their own local name for a plant or mosquito, what problems could happen? What kind of name could everyone in Kenya and the world agree on?' They then Think-Pair-Share their predictions and record one group prediction on a sticky note.	and say: 'Remember our first activity: the same organism had many names. Today we ask, how do we fix that?' Pose the question and give WAIT TIME of 10-15 seconds of silent thinking before any talking. Say: 'Write your own idea first. No one talks yet.' After pairing, cold-call 4 pairs (not volunteers) to share with: 'Tell us one problem your partner predicted.' Record predictions on the board under 'Our Predictions' without judging. Invite learners to add names in their home languages and say: 'Every name here is correct and valuable. Our job is to find a name all can share.' Provide vocabulary support for 'genus', 'species' and 'nomenclature' on a word wall in English and Kiswahili.	with Think-Pair-Share: writing a prediction first makes learners commit to ideas about naming problems, so evidence gathered later either confirms or challenges their thinking. Think-Pair-Share gives every learner, including quieter ones, a voice and surfaces the idea that common names vary by place and language.	of group work. Look for: learners who write a prediction before talking; predictions that mention confusion, mistakes in medicine/farming/pest control, or difficulty sharing research; learners who contribute local-language names. Note on a class list who mentions 'one agreed name' (ready) and who does not yet (needs support).
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy scientific name binomial  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms:	(Approx. 25 minutes) Part A (8 min): Groups complete a Name Comparison Table using printed cards: organism, Kikuyu name, Luo name, Kiswahili name, English name, and a column 'Do all communities use the same name?' They also look at examples where one common name refers to different organisms (e.g. 'corn' or 'robin' differ by country; 'sukuma' or 'kunde' leaves may refer to different plants in different regions) and where two organisms share a name. Part B (10 min): Learners watch the video on binomial nomenclature (https://www.youtube.com/watch?v=vRCdOOOfuv8) or read the printed video summary cards if there is no electricity/internet,	Distribute the Name Comparison Table and say: 'Look for patterns. Where do the names match and where do they differ? Write what you see, not what you think.' Circulate and ask: 'Which names would a visitor from Mombasa recognise? Why is this a problem for a doctor or a farmer?' Before the video say: 'Listen for three things: who, how many parts, and how it is written.' Pause the video twice (about 2 minutes and 5 minutes) and ask: 'What rule did you just hear?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. For the Jigsaw, assign expert numbers 1-4, time the expert groups for 4 minutes, and say: 'You are responsible for making sure your teammates can	Jigsaw with peer teaching and data analysis: each learner holds a unique piece of the rules and must teach it, creating positive interdependence and deeper understanding. Comparing the name table first creates the need for the rules, so the video evidence is explained against a real problem, not memorised.	Observation of group work, completed worksheets (Name Comparison Table and video note sheet). Check: table shows at least 3 organisms with differing local names; note sheet states 'two parts: genus and species', 'genus starts with a capital letter', 'italics or underline', 'one universal name'; each Jigsaw expert correctly explains their rule and gives one correct example. Peer assessment: each learner rates a teammate's explanation 1-3 using a checklist (clear, correct example, answered a question).

taxonomy scientific name binomial	completing a note sheet: Who developed the system? How many parts does a name have? Which part is the genus? Which is the species? How is it written? What language is used? Part C (7 min): Jigsaw. Each member of the home group of 4 becomes an expert on one rule: (1) two parts, genus and species; (2) capital letter for genus, small letter for species; (3) italics or underlining each part separately; (4) Latin form and one name worldwide. Experts meet in expert groups with examples (Zea mays, Phaseolus vulgaris, Capra hircus, Anopheles gambiae), then return to teach their home group (peer teaching).	use your rule by themselves.' Check that experts write examples correctly, e.g. the genus capitalised: Zea mays, not zea Mays. Support learners who need extra help with a printed rule card and example in both English and Kiswahili.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy scientific name binomial  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: taxonomy scientific name binomial	(Approx. 20 minutes) Learners write a Claim-Evidence-Reasoning (CER) explanation in groups answering: 'Why do scientists need a scientific name for each organism even though people already have common names?' Claim: a one-sentence answer. Evidence: at least two pieces from the name table or video (e.g. maize has different names in different communities; Zea mays is the same in Kenya, Brazil and China). Reasoning: link the evidence to the idea that a universal, two-part, Latin-based name avoids confusion and allows scientists everywhere to share information. Next, a card-sorting task: groups receive 10 name cards, some written correctly (Zea mays, Capra hircus) and some incorrectly (zea Mays, Phaseolus Vulgaris, Anopheles gambiae with no italics/underline, a one-word name). They sort into 'Correct' and 'Incorrect', correct the errors, and explain which rule was broken.	Model a weak and strong CER on the board using a different example: 'Claim: Mbuzi is not a good name for scientists. That alone is not enough. What evidence do we need?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners for evidence. Say: 'Reasoning is the because. Why does that evidence support the claim?' While groups write, circulate and ask: 'Which rule does Zea mays follow? Which does Phaseolus Vulgaris break?' For the card sort, give a 6-minute timer and say: 'Argue with evidence from the rules, not just opinion.' For the debate, divide the room into two sides ('keep local names' / 'use scientific only'), allow 2 minutes of preparation, and then cold-call 4 learners in total. Guide to a consensus: local names are valuable culturally and in communities; scientific names are essential for universal scientific communication. Praise respectful listening: 'Thank you for building	Claim-Evidence-Reasoning with card sorting and structured debate: CER forces learners to connect observations to the concept of a universal name; card sorting makes the rules concrete through error-spotting; the debate develops argumentation and respect for different viewpoints, linking back to the original phenomenon where everyone was 'correct' by their own criteria.	Group presentations, worksheets and observation. Look for: CER with a clear claim, at least two pieces of correct evidence, and reasoning that mentions 'universal' or 'avoids confusion'; card sort accuracy of at least 8 out of 10 with correct identification of the broken rule; debate contributions that use evidence and respectful language. Short quiz (3 questions, 3 minutes): (1) Write the scientific name of maize correctly. (2) Name the two parts of a scientific name. (3) Give one reason for using scientific names. Record who scores 3/3, 2/3 or below for follow-up.

	Finally, a short class discussion and debate: 'Should we stop using local names now that we have scientific names?' Learners take a position and give a reason, practising respect for diverse opinions.	on Wanjiru's point.' Probe misconceptions, e.g. 'Latin names are only for plants' or 'scientific names replace local names'.		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy scientific name binomial  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy scientific name binomial	(Approx. 8 minutes) Learners return to the Driving Question Board. In groups they write on sticky notes: (a) one question that has now been partly answered (e.g. 'Why do different communities have different names?' and 'Is there one name everyone can use?') and place it in the 'Answered / Partly Answered' section with a short answer; (b) one new question raised by this lesson, such as 'How do we decide the genus and species of an organism we find in our school compound?', 'Who gives the scientific names?', or 'How can we apply these naming rules to organisms found in our environment?'. Learners then do a quick Gallery Walk, reading other groups' notes and placing a tick next to a question they also wonder about.	Point to the DQB and say: 'Look at the questions from our first lesson. Which ones can we now answer using our evidence?' Give WAIT TIME of 10-15 seconds before groups write. Say: 'Your new question should start with how, why or what if, and should be something we can investigate.' Cold-call 3 groups to read a new question aloud and ask: 'What evidence led you to this question?' Cluster the questions and highlight the one that leads into Lesson 6: applying the rules to local organisms. Say: 'Next lesson you will become namers of organisms in our own environment.' Move answered questions to the 'Answered' section and thank groups for questions.	Driving Question Board with Gallery Walk: the board shows learners how their understanding grows, links each lesson to the driving question, and gives them ownership of the next investigation. The Gallery Walk exposes learners to peers' thinking and encourages shared curiosity.	Exit tickets and observation. Exit ticket (last 2 minutes): 'One thing I learned about scientific names today is... One question I still have is...'. Look for: correct mention of two parts/genus and species or universal use; questions that link to using the rules on local organisms. Sort tickets into 'secure', 'partly secure' and 'needs reteaching' for starting Lesson 6.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy scientific name binomial	(Approx. 15 minutes) Learners revise the class classification model, a large chart started in earlier lessons showing how organisms are grouped. In groups they add a new 'Naming Layer': for each of the eight original organisms (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito), they attach a card with the common names (Kikuyu, Luo, Kiswahili, English) and the correct scientific name where known (e.g. Capra hircus,	Display the original chart and say: 'At the start, our groups sorted by habitat, colour and usefulness, and nobody could agree. What does our model need so everyone can talk about the same organism?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. Provide the name cards and say: 'Write the scientific name exactly as the rules say. Check each other.' Circulate with a checklist: capital letter on genus, small letter on species, underlined/italic, two	Model revision: updating the class model with a universal naming layer lets learners see how a new idea resolves the original puzzle, where everyone was correct but could not communicate. Peer feedback on name writing reinforces the rules and builds self-efficacy.	Taxonomic hierarchy diagrams / classification card activities, peer assessment and observation. Look for: every organism has a two-part name with correct capitalisation and italics/underline; the model shows common names linked to one scientific name; revision note says that scientific names give one universal name that avoids confusion; peer checklist is accurate. Use results to plan Lesson 6 grouping and for the final classification challenge later in the

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy scientific name binomial	Gallus gallus domesticus, Zea mays, Phaseolus vulgaris, Anopheles gambiae; for mushroom, algae and bacteria the teacher provides example genus names such as Agaricus bisporus, Spirogyra, Escherichia coli). They draw a simple diagram of a scientific name labelling genus and species, and write a revision note: 'What our model could not explain before / what it explains now.' Groups share revisions with the class and give peer feedback on the correctness of the written names (capital letter, italics/underline).	parts. Ask: 'What part of the model still can't be answered?' Elicit that we know how names are written but not yet how to apply the rules to local organisms and assign them to kingdoms or other groups. Close with: 'Next lesson: How can we apply these naming rules to organisms found in our environment?' Ask learners to bring one safe, locally available plant leaf or picture from home or the school compound (not mushrooms or unknown moulds) and to observe road safety if collecting outside school.		sequence.
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D. TEACHER REFLECTION

Did learners make the link between the confusing common names in the opening phenomenon and the need for one universal scientific name? Which rule of binomial nomenclature (capital letter on genus, italics/underline, two parts, Latin form) was hardest, and which Jigsaw experts needed more support? Did the debate honour local-language names while valuing scientific names? Were all learners able to take part despite limited internet or electricity and the language of instruction? How might I adjust grouping, printed resources or vocabulary support in Lesson 6, when learners apply the naming rules to organisms in their own environment?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We compared the common names of organisms such as maize, beans, goat and mosquito in Kikuyu, Luo, Kiswahili and English and found they differ between communities. We watched a video and read about the two-part naming system and its rules.
What did I learn?	Scientific names give every organism one universal name that scientists everywhere can use. A scientific name has two parts, the genus (capital letter) and the species (small letters), written in Latin form and in italics or underlined separately, e.g. Zea mays.
How does this explain the phenomenon?	Common names vary with language and community, which causes confusion when sharing information, so a universal binomial name is needed. Because every organism has just one agreed scientific name, people everywhere can identify, study and communicate about it in the same way, which helps solve the puzzle from our original sorting activity.

LESSON 6 (40 minutes (single lesson; extend to a double period if the school walk is included)): Applying Binomial Nomenclature: Naming Our Local Organisms the Same Way Everywhere

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the rules of binomial nomenclature to correctly write scientific names of common Kenyan organisms, and use this to explain how scientific names let people everywhere identify the same organism, resolving the puzzle of different groupings and local names (Kikuyu, Luo, Kiswahili, English) seen in the anchoring phenomenon.
Knowledge	By the end of the lesson the learner should be able to: c) describe the principles of binomial nomenclature in classification of living organisms. Learners recall that a scientific name has two parts, genus (capital first letter) and species (lower case), written in italics when typed or underlined separately when handwritten.
Skills	Constructing explanations (NGSS SEP 6): using Claim-Evidence-Reasoning to explain why scientific names are universal. Analysing and interpreting data: checking names against the rules. Obtaining, evaluating and communicating information: using a video or printed photographs of local organisms. Developing and using models: adding naming rules to the class classification model.
Attitudes	Respect: the learner appreciates diverse opinions while discussing the principles of binomial nomenclature with peers and values local-language names. Responsibility: the learner observes safety precautions while naming common organisms in the local environment. Communication and Collaboration: contributes to group decision-making. Safety and Security (PCI): observes road safety when exploring the school environment. Environmental Conservation: appreciates that a shared naming system supports biodiversity conservation.
Key Inquiry Question	How are living organisms classified? Can characteristics shared by organisms help us group them into larger categories?
Purpose in Storyline	Lessons 1-5 showed that groups sorted organisms by different criteria and that local names differ (for example maize is 'mahindi' in Kiswahili, 'mbembe' in Kikuyu, 'oganda' in Luo). In Lesson 6 learners use the binomial rules to give each organism one name recognised worldwide. This resolves the naming half of the puzzle and raises the next question: once organisms share a genus, how do genera group into bigger categories (family, order, class, phylum, kingdom)?
Safety Notes	If walking around the school compound: go in groups of 5-6 with a team leader, stay inside the school fence, use only paths away from vehicles and the school gate, and never cross a road. Do not touch mushrooms, unknown plants, insects or stagnant water; observe, photograph or sketch only. Wash hands after outdoor work. If there is no electricity, use printed photographs and a teacher-read narration instead of the video. Do not culture bacteria or mould.

B. LESSON OVERVIEW

Learners return to the original puzzle: the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito were grouped and named differently by different groups. In this lesson they predict how a Luo farmer in Kisumu and a Kikuyu trader in Nairobi could be sure they mean the same maize or mosquito. They watch a short video (or use printed photographs and a teacher narration) showing Linnaeus' two-name system and examples of Kenyan organisms, then check and correct scientific names on cards, write names correctly using the rules, and explain with Claim-Evidence-Reasoning why scientific names are universal. They add a 'Naming' section to the Driving Question Board and revise the class model of classification.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy binomial nomenclature genus  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: taxonomy binomial nomenclature genus	(6 min) Each learner gets a card with one organism from the original phenomenon set (goat, chicken, maize, mushroom, algae, bacteria, bean plant, mosquito) and its names in Kikuyu, Luo, Kiswahili and English where known. Learners individually write a prediction: 'If a scientist in Nairobi and one in Japan both want to write about this organism, what name could they both use so nobody is confused? What might that name look like?' They then Think-Pair-Share with a partner and sketch how they think the name should be written.	Display the organism cards on the board. Say: 'Remember our first day, when the groups sorted these organisms differently and gave them different names. Today we fix the naming problem.' Ask: 'Mahindi, mbembe, oganda, maize. Which is the correct name?' WAIT TIME 10-15 seconds before taking answers. Give 2 minutes of silent writing, then 2 minutes pair talk. Cold-call 4 pairs (not volunteers) to share predictions and record every prediction on the board without judging. Circulate and listen for ideas such as 'use English', 'use Latin', 'use two words'.	Predict-Observe-Explain combined with Think-Pair-Share: surfacing learners' initial ideas about a universal name creates a purpose for watching the video and allows learners to compare their later observations with their predictions.	Oral questioning and observation of pair talk. Indicators: learner says that one common name for all people is needed; learner suggests a name of more than one word or in a shared language; learner notes that local names differ. Tally on a checklist how many pairs mention 'one agreed name' versus 'local names only'. Use this to decide how much recap of Lessons 4-5 is needed.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy binomial nomenclature genus  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms:	(12 min) Learners watch the video (about 5 minutes) on binomial nomenclature, or use printed photographs and a teacher narration if there is no power. In groups of four they use a Jigsaw: each member becomes an expert on one rule: (1) two parts: genus and species, (2) genus starts with a capital, species with a small letter, (3) italics when printed, underline each part separately when handwritten, (4) names are in Latin/Latinised and the same worldwide. Experts then teach their rule in the home group (Peer teaching) and the group completes a table with the scientific names of Zea mays (maize), Phaseolus vulgaris (bean), Capra hircus (goat), Gallus gallus domesticus	Before the video say: 'Watch for two things: how many parts the name has, and how it is written.' Pause the video twice and ask: 'What did you notice about the first word?' WAIT TIME 10-15 seconds, then cold-call 3 learners each time. Without electricity, read the narration aloud slowly and show printed photographs. Assign jigsaw rules by number, giving 3 minutes for expert preparation and 5 minutes for teaching in home groups. Circulate with a checklist. Provide vocabulary support: write 'genus' (jenasi), 'species' (spishi) and 'scientific name' (jina la kisayansi) on the board. Remind outdoor groups: 'Stay together, stay inside the fence, no touching unknown plants, and no going	Jigsaw with Peer teaching: each learner owns one piece of the evidence and must teach it, so every learner interacts with all four rules. Card sorting (matching scientific names to organism cards) turns the abstract rules into a concrete task that links back to the original phenomenon.	Observation of group work and classification card activity. Indicators: expert accurately states their rule without reading; group correctly matches at least 4 of 5 scientific names to organisms; learners note the genus has a capital letter and the species does not; learners underline each part separately in handwritten names. Record names of learners who teach their rule correctly. Peer assessment: each group member ticks 'explained clearly' or 'needs help' for the expert in their group.

taxonomy binomial nomenclature genus	(chicken) and <i>Anopheles gambiae</i> (mosquito), matching them to the organism cards. Where possible, a group of learners goes with the teacher to the school garden to observe a maize or bean plant and sketch it with its correct scientific name, observing road safety rules.	near the gate or road.'		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy binomial nomenclature genus  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy binomial nomenclature genus	(10 min) Learners complete a worksheet 'Fix the Names': a list of incorrectly written names (for example 'zea Mays', 'Capra Hircus', 'phaseolus vulgaris' in plain print, 'Anopheles Gambiae') to correct, then write scientific names for two new local organisms supplied by the teacher (for example <i>Mangifera indica</i> , mango; <i>Solanum lycopersicum</i> , tomato; <i>Brassica oleracea</i> , sukuma wiki/kale). Then they write a Claim-Evidence-Reasoning response to: 'Why is Zea a better name than mahindi, mbembe, oganda or maize for scientists in Kenya and around the world?' Finally, in a short discussion and debate, learners respond to the statement: 'Local names are not important when we have scientific names.'	Model the first correction on the board: 'zea Mays' and ask 'What is wrong? Look at the rules.' WAIT TIME 10-15 seconds. Provide the CER frame: 'Claim: ... Evidence: from the video and cards, ... Reasoning: because ...'. Ask learners to work for 5 minutes. Then cold-call 4 learners to share a claim and evidence and ask 2 others: 'Do you agree or disagree and why?' In the debate, require respectful language: 'I respectfully disagree because...' and affirm that local names such as mbembe or oganda carry cultural and practical value, while scientific names allow worldwide communication. Close with: 'So how does a scientific name help us in the original puzzle?'	Claim-Evidence-Reasoning requires learners to link rules and examples to the claim that scientific names are universal, building scientific explanation. Discussion and debate develops respect for diverse views and prevents the misconception that scientific names replace local names.	Worksheet marking and short quiz. Indicators: learner corrects at least 4 of 5 names, correctly using capital genus, lower-case species and underlining or italics; CER includes a claim (scientific names are universal), evidence (same two-part name used worldwide, differing local names) and reasoning (so everyone can identify the same organism). Look for common errors: capitalising the species, underlining both words as one, and writing common names in italics. Note learners to revisit in the next lesson.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy binomial nomenclature genus	(5 min) Learners return to the Driving Question Board. In pairs they write one new sticky note: either an answer to an earlier question now resolved (for example 'Why do people call the same plant different names?') with evidence from today, or a new question (for example 'Do organisms with the same genus belong to bigger groups?', 'Are the five kingdoms named in the same way?'). They place the notes on the board under 'Naming' or	Say: 'Look at our board. Which question can we now answer using today's evidence?' WAIT TIME 10-15 seconds. Invite 3 pairs (cold-call) to read and place notes, and ask each: 'What evidence helped you answer it?' Move answered questions to the 'Answered' column and circle one new question to guide Lesson 7: 'Can organisms with similar features be grouped into larger categories?' Ask learners to copy the new question into their notebooks.	Driving Question Board: gives learners ownership of the inquiry by linking evidence to earlier questions and generating new questions that motivate the next lesson.	Observation of the quality of notes. Indicators: learners cite today's evidence (rules, Zea mays example) when answering; new questions are testable and connect to the hierarchy of taxonomic units; every pair posts at least one note. Exit ticket: 'Write the scientific name of one organism correctly and one reason scientific names are useful.'

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy binomial nomenclature genus	'Grouping'.			
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy binomial nomenclature genus  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy binomial nomenclature genus	(7 min) Learners revise the class model of classification (a chart that began with sorting criteria and the need for a universal system). In groups they add a 'Naming' panel: a diagram showing one organism (for example maize) with its local names branching to one scientific name (<i>Zea mays</i>), annotated with the four rules. They then do a Gallery Walk to view other groups' models, leave one 'star' and one 'wish' on a sticky note, and revise their own model if necessary.	Say: 'Our model must now answer: how can everyone name and identify the same organism? Add what you learned today.' Provide chart paper and markers. Circulate and prompt: 'Where on your model do we see the genus and the species? Which rule is easiest to forget?' WAIT TIME 10-15 seconds after each question. During the Gallery Walk ask 3 learners to explain one feature on a model they visited. End with a final checkpoint: 'Is anything still missing from our model?' Collect learners' notes about what they still wonder (hierarchy, kingdoms).	Model revision with Gallery Walk and peer assessment: learners make their thinking visible, compare models and refine them using evidence, showing how naming rules fit into the larger classification system.	Taxonomic hierarchy diagram and group presentation checks. Indicators: model shows several local names leading to one scientific name; rules are correctly labelled; scientific names are written correctly; peer feedback is specific (for example 'species should start with a small letter'). Teacher records which groups need support with writing conventions before the final classification challenge.

D. TEACHER REFLECTION

Which learners still capitalise the species name or fail to underline separately, and what additional practice with examples like *Zea mays* and *Phaseolus vulgaris* do they need? Did the Jigsaw give every learner a chance to speak, or did a few dominate? Was the video or the printed photograph option more effective given the electricity situation? Did learners show respect for local-language names during the debate? How well did the DQB notes show readiness to move from naming to the hierarchy of taxonomic units in Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the same organism (such as maize, goat or mosquito) has different local names but only one two-part scientific name (for example <i>Zea mays</i> , <i>Capra hircus</i> , <i>Anopheles gambiae</i>), written with a capital genus, a lower-case species, and italics or underlining.
What did I learn?	We learned the principles of binomial nomenclature: two names (genus and species), correct capitalisation, italics or underlining, and the use of Latin so the name is the same worldwide.
How does this explain the phenomenon?	We can explain that scientific names give a universal way of identifying organisms, so people who use different local names (Kikuyu, Luo, Kiswahili, English) can still be sure they are talking about the same organism, which solves part of our driving question.

LESSON 7 (80 minutes (double lesson: 2 x 40 minutes)): Five Kingdoms of Living Organisms

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use pictures, charts, specimens and videos to investigate one kingdom each, identify its characteristics (cell type, cell organisation, mode of nutrition, examples), and teach peers in jigsaw groups so that they can explain how organisms are grouped into five kingdoms using shared characteristics.
Knowledge	Specific Learning Outcomes: By the end of the lesson, the learner should be able to (a) classify organisms into the five taxonomic Kingdoms (Monera, Protocista, Fungi, Plantae, Animalia), (b) describe the general characteristics of the five kingdoms, and (c) give local examples for each kingdom, e.g. bacteria in sour milk (maziwa lala), Spirogyra and algae from Lake Victoria, mushrooms and bread mould, maize and bean plants, goat, chicken and mosquito.
Skills	NGSS Science and Engineering Practices: Analyzing and Interpreting Data (comparing kingdom characteristics in a table), Obtaining, Evaluating and Communicating Information (jigsaw teaching), Engaging in Argument from Evidence (Claim-Evidence-Reasoning), Developing and Using Models (revising the class classification model). Core competencies: Communication and Collaboration - the learner contributes to group decision-making while exploring the local environment/print media/non print media to observe and classify organisms.
Attitudes	Values: Respect - the learner appreciates diverse opinions while discussing with peers. Responsibility - the learner observes safety precautions when handling specimens. PCIs: Safety and Security (road safety if specimens are collected outdoors); Environmental Conservation - the learner appreciates the need to conserve biodiversity.
Key Inquiry Question	How are living organisms classified?
Purpose in Storyline	In Lessons 1-6 learners found that sorting by colour, habitat or usefulness, and local names in Kikuyu, Luo, Kiswahili and English, cannot give everyone the same system, and they met the hierarchy and binomial naming. Lesson 7 gives the first large-scale scientific grouping: the five kingdoms based on shared structural and nutritional characteristics. It resolves the puzzle of the goat, chicken, maize, mushroom, algae, bacteria, bean plant and mosquito by placing each in a kingdom, and it raises the next question: Can we use the characteristics we have discovered to classify unfamiliar organisms into the correct kingdom?
Safety Notes	Use printed photographs, charts and prepared slides where possible. Do NOT culture unknown bacteria or mould. Handle mushrooms, bread mould and pond water only in sealed containers or under a hand lens, and never taste specimens. Wear gloves or wash hands with soap after handling. Use only locally available safe specimens (maize and bean leaves, bought mushroom, dry mould on old bread sealed in a clear bag, Spirogyra in a jar, preserved insect). If collecting outdoors, follow road safety rules and stay in pairs.

B. LESSON OVERVIEW

Learners return to the original phenomenon: goat, chicken, maize, mushroom, algae, bacteria, bean plant and mosquito, which different groups sorted differently and named differently in Kikuyu, Luo, Kiswahili and English. They first predict how a scientist would group these eight organisms, then investigate one kingdom per expert group using pictures, charts, specimens or videos (printed photographs if electricity is unreliable), recording cell type, cell organisation, mode of nutrition and examples. Jigsaw groups then teach one another, use Claim-Evidence-Reasoning to place the eight organisms into kingdoms, update the Driving Question Board, and revise the class model. The lesson ends by posing the next question: can we classify unfamiliar organisms into the correct kingdom?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Autotrophs and Heterotrophs (Flexbook) Source: CK-12  Search ARES for similar readings	Learners see the eight phenomenon organisms (printed photos on the board or desk cards): goat, chicken, maize, mushroom, algae, bacteria, bean plant, mosquito. Individually they write a prediction: 'If scientists worldwide used the same system, how many big groups would these eight fall into, and which organisms would share a group? What characteristic would they use?' They Think-Pair-Share with a partner, then record a group prediction in their notebooks.	Display the cards and say: 'Remember our first lesson, when your groups sorted these by colour, habitat and usefulness and every group was right, yet we could not share one system. Today we find out what scientists use instead.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before pairing. Say: 'Turn to your partner and share your prediction and your reason.' Cold-call 4 pairs: 'What did your partner predict, and what characteristic did they use?' Record predictions on the board without judging; ask: 'Which characteristic could be checked by anyone, in any country?' Circle ideas such as cell structure and feeding.	Predict-Observe-Explain with Think-Pair-Share: committing to a prediction makes learners aware of their existing ideas (often colour, size, habitat), so they notice when evidence contradicts or supports them later in the lesson.	Oral questioning and observation: note how many pairs use observable structural or feeding characteristics versus appearance or usefulness. Indicator: at least half of pairs can name one characteristic that is not colour, size or usefulness. Record misconceptions (e.g. 'mushrooms are plants because they grow in soil') for revisiting in the Explain phase.
Observe Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Autotrophs and Heterotrophs (Flexbook) Source: CK-12  Search ARES for similar readings	Jigsaw Step 1 (expert groups, about 20 minutes): The class forms five expert groups, one per kingdom (Monera, Protocista, Fungi, Plantae, Animalia). Each group receives a kingdom evidence pack: printed photographs or charts (the 'Classification' resource/textbook pages shared in groups), local specimens or prepared slides, and a video clip if electricity is available. Examples: Monera - photos of bacteria, cyanobacteria, and maziwa lala (fermented milk) as a product of bacteria; Protocista - Spirogyra or green pond water from a jar, Amoeba and Paramecium prepared slides, algae from Lake Victoria; Fungi - bought mushroom, sealed bread mould, yeast used for mandazi	Distribute packs and state: 'You are now the class experts on one kingdom. In 20 minutes each of you must be able to teach someone who has seen nothing about your kingdom.' Review safety: 'No tasting. Wash your hands after handling the mushroom and mould. Mould stays sealed in its bag.' Circulate to every group at least once, asking: 'What evidence on this card tells you the cell type?' and 'How does this organism get its food?' After a question WAIT 10-15 seconds before prompting. Prompt groups with unclear photos: 'What can you see with the hand lens that the photograph cannot show?' Support vocabulary with bilingual terms (e.g. fungi - kuvu, bacteria - bakteria, plants -	Group investigation with evidence from multiple sources (specimens, pictures, charts, video) builds the shared categories for the kingdom data. The Kingdom Data Card forces learners to move from looking at organisms to identifying the underlying characteristics used in classification. Making every member an expert (individual accountability) prepares for peer teaching.	Observation of group work with a checklist: (1) every member contributes data, (2) cell type, organisation and nutrition are recorded correctly for at least 3 examples, (3) learners use terms such as prokaryotic, eukaryotic, autotroph, heterotroph. Oral questioning during circulation. Indicator: each group's data card has no blank cells and includes one Kenyan example. Cold-called learners explain the data using evidence from the specimen or picture, not guesses.

	dough, under a hand lens; Plantae - maize, bean plant, sukuma wiki, moss/fern; Animalia - goat, chicken, mosquito (preserved or pictures). Using a Kingdom Data Card, they record: cell type (prokaryotic or eukaryotic), cell organisation (unicellular, multicellular, colonial), cell wall present or absent and its material, mode of nutrition (autotrophic, heterotrophic, saprophytic, parasitic), and at least four examples, including one Kenyan example and its local name. Each member must be able to explain all the entries.	mimea, animals - wanyama). Check that each member can summarise by cold-calling 2 learners per group: 'Explain your kingdom's nutrition in your own words.' Give a 2-minute warning.		
Explain Phase  VIDEO: Cells and organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Prokaryotes and eukaryotes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Jigsaw Step 2 (teaching groups, about 20 minutes): Learners regroup into mixed jigsaw groups of five, each containing one expert from each kingdom. Each expert teaches their kingdom in 3 minutes using the data card and a specimen or picture while others complete a master comparison table with columns: kingdom, cell type, cell organisation, cell wall, nutrition, examples. Learners then use Claim-Evidence-Reasoning on the eight phenomenon organisms: 'Claim: the mushroom belongs in Kingdom _____. Evidence: _____. Reasoning: because _____. They must resolve the debate about the mushroom (grows in soil but is heterotrophic, cell wall of chitin, not a plant) and algae (not a plant despite being green, in Protocista). They then do a card sort of the eight organisms into the five kingdoms and justify disagreements politely, listening to differing views.	Say: 'Experts, you are now the teachers. Listeners, you will complete the table and ask at least one question.' Time each 3-minute turn. During teaching circulate and listen for accuracy. Pose: 'The mushroom grows from the soil next to the maize plant. Why is it not a plant?' WAIT 10-15 seconds, then ask for a Claim-Evidence-Reasoning response from 3 cold-called learners. Ask: 'Algae are green and live in water like Spirogyra in Lake Victoria. Does being green make them plants? What evidence would you need?' Correct errors with questions instead of answers: 'Which column of your table contradicts that?' Highlight respect for varying opinions: 'Challenge the idea, not the person.' Ask: 'How does this single system fix the problem we had when the goat, chicken and mosquito had different names and groupings?' Cold-call 4 learners for final placement of the eight organisms.	Jigsaw and peer teaching require learners to explain concepts to others, strengthening understanding. Claim-Evidence-Reasoning ties each placement to observable characteristics, shifting from opinion (colour, usefulness) to evidence-based argument. Discussion and debate on borderline cases (mushroom, algae) highlights why shared characteristics, not appearance, define kingdoms. Card sorting makes the final classification concrete.	Peer assessment: listeners rate their expert (accurate, clear, used evidence) with a tick-scale and one comment. Worksheet: completed comparison table checked for accuracy. Classification card activity: indicator - at least 7 of the 8 organisms placed correctly with a CER statement mentioning cell type or nutrition. Short quiz (3 questions, e.g. 'Which two kingdoms are made of prokaryotic cells?' Actually only one: Monera; 'Name an autotroph from Kingdom Protocista'; 'Why is a mushroom not a plant?'). Record learners who still rely on colour or habitat.
Driving	Learners gather at the DQB (chart	Say: 'Look at our Driving Question	The Driving Question Board	Observation of DQB contributions

Question Board (DQB) Creation  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Autotrophs and Heterotrophs Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	paper or wall space). Using sticky notes they (1) add answers or evidence to earlier questions, e.g. 'How can we organise living organisms so everyone can study them the same way?' with 'Five kingdoms by cell type, organisation and nutrition'; (2) add new questions arising from the lesson such as 'Where do viruses belong?', 'Can we classify an organism we have never seen?', 'Why are some protocists plant-like and others animal-like?', 'How are kingdoms divided into smaller groups?'. Groups place the Kenyan examples they found next to the matching questions. A Gallery Walk lets groups read and star the questions they consider most important.	Board. Which of our earlier questions can we now answer with evidence from today?' WAIT 10-15 seconds. Cold-call 3 learners to move a note and justify it. Ask: 'What new questions has the five-kingdom system raised for you?' Give 3 minutes for writing, then a 5-minute Gallery Walk: 'Walk around, star two questions you think we must solve next.' Highlight the question for the next lesson: 'Can we use the characteristics we have discovered to classify unfamiliar organisms into the correct kingdom?' Link to Kenyan context: 'If a scientist at KEMRI or the National Museums of Kenya finds an unknown organism in the Rift Valley, how could the five characteristics help?'	makes learning visible: learners connect new evidence to the central problem of a shared classification system and see how each lesson builds on the previous ones. Gallery Walk lets learners read peers' ideas and collectively prioritise questions, giving ownership of the next inquiry.	and sticky notes. Indicator: at least one note per group links kingdom characteristics (not appearance) to the driving question; new questions show curiosity about unfamiliar or borderline organisms. Exit ticket (one slip each): 'Write one characteristic used to group organisms into kingdoms and one question you still have.' Collect for planning the next lesson.
Model Building Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Autotrophs and Heterotrophs (Flexbook) Source: CK-12  Search ARES for similar readings	Groups revise their class classification model, started in earlier lessons (a poster or flow diagram of the hierarchy from kingdom downward and the binomial name idea). They add a new layer: a five-kingdom diagram or key showing cell type (prokaryotic vs eukaryotic), cell organisation (unicellular vs multicellular), and mode of nutrition (autotrophic vs heterotrophic), with each of the eight phenomenon organisms placed in the correct kingdom, including a Kenyan example for each. They annotate what changed from their original sorting (by colour, habitat, usefulness) and write a 2-3 sentence explanation: 'Our model now explains why everyone can classify the same way because...'. Groups present briefly and peers give feedback.	Say: 'Open your class model from Lesson 6. Add today's evidence. What must change so a learner in Kisumu, Nairobi or Mombasa gets the same classification as you?' Allow 10 minutes for revision, circulating with: 'Which characteristic did you use to place the algae, and why that one?' WAIT 10-15 seconds after each question. Choose 3 groups for 1-minute presentations; ask the class to give one 'glow' (strength) and one 'grow' (suggestion). Ask a group with an error: 'What evidence in the table could fix this placement?' Close with a 2-question mini-quiz by show of cards (A-E for kingdoms): 'Cell wall of chitin, heterotroph, eukaryotic?' (Fungi) 'Prokaryotic, no nucleus?' (Monera). Preview next lesson: unfamiliar organisms for the final classification challenge. Homework: find two organisms at home or on the way to school	Model revision lets learners integrate new evidence into a cumulative explanation, compare it with their earlier ideas, and recognise how scientific classification improves on local, appearance-based grouping. Presenting and peer feedback strengthen self-efficacy and communication.	Taxonomic hierarchy diagram and model check: indicators - all five kingdoms included, each defined by at least two shared characteristics, all eight organisms placed correctly, Kenyan examples included. Group presentation rated with a simple rubric (accuracy, evidence, clarity, teamwork). Short card quiz gives a quick class-level check; learners below 70% receive extra support with the table in the next lesson, in preparation for the final classification challenge.

		(observing road safety) and note their characteristics.		
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D. TEACHER REFLECTION

Which learners still sorted by colour, habitat or usefulness instead of cell type and nutrition, and what evidence changed (or did not change) their thinking? Were all expert groups able to access reliable evidence (specimens, photographs, charts) given electricity and resource limits? Did the mushroom and algae debates help learners see why shared characteristics matter? Which jigsaw experts taught accurately, and which kingdom was taught least clearly? How did vocabulary support and local language names help or confuse? What adjustments are needed to prepare learners to classify unfamiliar organisms in the next lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did you observe about the cells, organisation and feeding of organisms in each of the five kingdoms (for example bacteria in maziwa lala, Spirogyra, mushroom, maize and bean plants, goat, chicken and mosquito)?
What did I learn?	What characteristics do scientists use to group organisms into the five kingdoms, and how are they different from colour, habitat or usefulness?
How does this explain the phenomenon?	How does the five-kingdom system solve the problem of organisms having different groupings and names in Kikuyu, Luo, Kiswahili and English, and what question does it raise about classifying unfamiliar organisms?

LESSON 8 (80 minutes (double lesson)): Classification into the Five Kingdoms and Importance of Taxonomy: Our Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To bring together all evidence from Lessons 1-7 so learners can classify unfamiliar organisms into the five kingdoms using observable evidence, defend their decisions, and give a final explanation of how living organisms are systematically organised, named and identified.
Knowledge	By the end of the lesson the learner should be able to: a) explain the significance of classification of living organisms in nature; b) illustrate the hierarchy of the major taxonomic units of classification; c) describe the principles of binomial nomenclature in classification of living organisms; d) classify organisms into the five taxonomic Kingdoms; e) appreciate the importance of taxonomy in biodiversity conservation.
Skills	Classifying using observable characteristics; Constructing explanations (CER: Claim-Evidence-Reasoning); Engaging in argument from evidence; Developing and revising models; Communication and Collaboration: contributing to group decision-making while classifying organisms; Self-efficacy: presenting a taxonomic hierarchy chart to peers with confidence.
Attitudes	Respect: appreciating diverse opinions (including different local names and different first classifications) while discussing and defending classification choices. Responsibility: observing safety precautions when handling specimens and naming organisms with binomial nomenclature. Environmental Conservation (PCI): appreciating the need to conserve biodiversity. Safety and Security (PCI): observing road safety whenever exploring the local environment to observe organisms.
Key Inquiry Question	How are living organisms classified? Final driving question: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?
Purpose in Storyline	Lesson 8 is the culminating lesson. Learners return to the Lesson 1 puzzle (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito sorted by where they live, colour, size or usefulness, with names differing in Kikuyu, Luo, Kiswahili and English). They now apply the five-kingdom system, the taxonomic hierarchy and binomial nomenclature to unfamiliar organisms, revise their original classification, complete the Driving Question Board, and write the final explanation: living organisms can be systematically organised (classification into kingdoms and a hierarchy), named (binomial nomenclature) and identified (observable characteristics), through the work of taxonomy.
Safety Notes	Use printed photographs and profile cards; do not culture or open any unknown bacteria or mould. Use only prepared slides for Monera, Protocista and Fungi, and handle microscopes and slides with both hands. Wash hands if any real specimen (e.g. bean plant, maize leaf, mushroom from the market) is handled; do not taste any specimen. If learners go outside for the optional observation, they walk in pairs, stay inside the school compound, and observe road safety. Share textbooks in groups. Offer vocabulary support with bilingual terms (e.g. uyoga = mushroom, mwani = algae, bakteria = bacteria).

B. LESSON OVERVIEW

This final lesson has learners apply everything they have learned to a new challenge, the Final Classification Challenge. Learners first predict how they would now sort the Lesson 1 organisms (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito). They then gather evidence from unfamiliar organism profile cards (e.g. Spirogyra from Lake Victoria shores, Escherichia coli, Rhizopus on stale mandazi/bread, Agaricus (mushroom), Zea mays, Tilapia (Oreochromis niloticus), Anopheles mosquito, Amoeba, Acacia from the Rift Valley), supported by the classification video or reading. A Jigsaw expert-group structure and card sorting lead to groups defending their classifications with Claim-Evidence-Reasoning. In the Explain phase, learners connect classification, naming (binomial nomenclature) and taxonomy as one system and discuss its importance for biodiversity conservation. The class completes the Driving Question Board, then learners compare their Lesson

1 model with a final revised model and write the Final Explanation answering: How can living organisms be systematically organised, named and identified?

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Prokaryote Classification (Flexbook) Source: CK-12  Search ARES for similar readings	1. Learners recall the Lesson 1 puzzle. The teacher displays the Lesson 1 picture set (or printed photos) of goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito, together with the groups' original sorting posters pinned on the wall. 2. Individually, learners write for 2 minutes in their notebooks: 'If we sorted these 8 organisms again today, which groups would I use, and what evidence would I use for each group?' 3. Think-Pair-Share: they pair up and compare predictions. 4. They also predict: 'Here are three organisms I have never shown you: a single-celled organism from a pond (Amoeba), a mould on bread (Rhizopus) and a tilapia from Lake Victoria. Which kingdom might each belong to and why?' Learners write a prediction with a confidence rating (1-3).	Say: 'On the wall are the posters you made in Lesson 1. Everyone was right by their own rule, yet nobody could agree. Today we finish solving that puzzle. Before we begin, I want your best thinking now.' Give 2 minutes of silent writing. Circulate and note 3 learners who still sort only by habitat, colour or usefulness and 3 who already use cell structure or mode of nutrition. Say: 'Turn to your partner, share in Kikuyu, Luo, Kiswahili or English if you wish, then give me the evidence, not just the answer.' WAIT TIME: after asking 'What kind of evidence would scientists everywhere agree on?', wait 10-15 seconds before taking responses. Cold-call 4 learners (including at least 2 who rarely speak) to share their partner's idea, not their own. Record emerging criteria on the board under 'Our evidence ideas' without correcting yet. Say: 'Keep your prediction sheet; we will return to it at the end.'	Think-Pair-Share combined with Predict-Observe-Explain: committing to a prediction in writing before new evidence surfaces prior ideas (habitat, colour, usefulness), so learners can see for themselves how their thinking has changed. Cold-calling on a partner's idea promotes active listening and Respect for diverse opinions.	Oral questioning and observation: Does the learner name at least one observable structural feature (e.g. presence of a nucleus, cell wall, chlorophyll, mode of nutrition) rather than only habitat/use? Tally on a class checklist: number of learners using (a) habitat/colour/use only, (b) some structural evidence, (c) kingdom-level criteria. Collect confidence ratings to compare with later performance.
Observe Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12	1. Learners are in Jigsaw home groups of 5, then split into expert groups. Each expert group receives a set of printed profile cards and photographs for ONE kingdom cluster: Group A: Monera (bacteria e.g. Escherichia coli, cyanobacteria) and Protoctista (Amoeba, Spirogyra, Paramecium); Group B: Fungi (Rhizopus on stale bread,	Say: 'You are now taxonomists working for the Kenya Wildlife Service and the National Museums of Kenya. You have unknown organisms; your job is to place each in the correct kingdom using evidence only.' Distribute cards and materials; display video or reading link/page number. Pause the video 3 times with prompts: 'What did you see that tells you the	Jigsaw + Card Sorting + Peer Teaching: each learner becomes responsible for evidence about one kingdom and teaches it, ensuring accountability. Card sorting makes the criteria for classification visible and testable. The evidence table links observable features to kingdom, helping learners shift from 'where it lives or how useful it is' to shared	Classification card activity and observation of group work. Indicators: (1) correctly places at least 7 of 9 unfamiliar organisms using stated evidence; (2) completes the evidence table with correct cell type and nutrition for each organism; (3) writes at least 3 binomial names with correct format (capital genus, lowercase species, italicised/underlined); (4)

Search ARES for similar videos EXERCISE: Prokaryote Classification Practice Source: CK-12 Search ARES for similar readings	Agaricus mushroom, yeast used in making mandazi/ujji); Group C: Plantae (Zea mays/maize, Phaseolus vulgaris/bean, sukuma wiki Brassica oleracea, Acacia, fern, moss); Group D: Animalia (goat Capra hircus, chicken Gallus gallus, tilapia Oreochromis niloticus, Anopheles mosquito); Group E: Mixed unfamiliar organisms (e.g. Euglena, Volvox, Nostoc, Penicillium, Spirulina from Lake Bogoria). 2. Each profile card lists observable evidence: cell type (prokaryotic/eukaryotic), unicellular or multicellular, cell wall presence/composition, chlorophyll, mode of nutrition, movement, reproduction. 3. Experts use the classification video (or the shared textbook reading if no electricity) and a prepared-slide or hand-lens observation station (where available) to fill an evidence table: Organism Cell type Unicellular/Multicellular Nutrition Cell wall Other features Kingdom? 4. Experts then return to home groups and teach their kingdom (Peer teaching). Home groups complete a card sort of all organisms into five kingdom envelopes. 5. Each group also arranges the taxonomic hierarchy cards (Kingdom, Phylum, Class, Order, Family, Genus, Species) for one organism, e.g. maize (Plantae > Magnoliophyta > Liliopsida > Poales > Poaceae > Zea > Zea mays), and writes its binomial name correctly (genus capitalised, species lowercase, italicised or underlined).	cell type?' WAIT TIME 10-15 seconds after each prompt. Circulate with a checklist; ask groups: 'What observable feature made you choose this kingdom? What would change your mind?' Redirect common errors: 'Is a mushroom a plant? What does it use for food? Does it have chlorophyll?' and 'Is Spirogyra an animal because it lives in water?' Use only prepared slides; remind learners about microscope handling and to wash hands after handling real specimens. During the home-group teaching, cold-call 5 learners to explain a kingdom they did not research themselves. Say: 'Remember: every voice counts; listen to the expert and ask a question if the evidence is not clear.' For binomial names, say: 'Show me how you write Zea mays; what rule did you follow?' Provide Kiswahili/vernacular terms alongside scientific names (e.g. mahindi - Zea mays, maharagwe - Phaseolus vulgaris, sukuma wiki - Brassica oleracea) so learners link local names to one universal name.	biological evidence.	can sequence the seven taxonomic ranks from kingdom to species correctly; (5) expert accurately teaches peers (peer assessment checklist: clear, accurate, used evidence). Note groups placing mushroom in Plantae or algae in Animalia for follow-up.
Explain Phase VIDEO: 24 Six Kingdoms Of Classification	1. Each home group selects 3 challenging organisms (e.g. mushroom, Spirogyra, bacteria) and prepares a Claim-Evidence-	Say: 'A claim without evidence is just an opinion. A scientist must show the evidence and explain the reasoning.' Model one CER aloud:	Claim-Evidence-Reasoning + Gallery Walk + Discussion and debate: CER forces learners to justify with observable evidence	Group presentations, peer assessment and oral questioning. Indicators: CER charts contain (a) a clear kingdom claim, (b) at least

<i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  EXERCISE: Classification of Organisms Practice Source: CK-12  Search ARES for similar readings	Reasoning (CER) statement on a chart: Claim (kingdom), Evidence (observable features from cards/slides/video), Reasoning (why those features define that kingdom, link to kingdom characteristics). 2. Gallery Walk: groups post charts; learners rotate every 3 minutes with sticky notes giving peer feedback: a star (strong evidence) and a question (unclear/disputed). 3. Class discussion and debate on a disputed case such as: 'Is algae (Spirogyra) a plant?' or 'Are mushrooms plants because they grow in soil?' Learners defend positions with evidence and revise where evidence demands. 4. Learners connect the three processes: classification (grouping using shared characteristics), nomenclature (one two-part scientific name like <i>Anopheles gambiae</i> understood by scientists worldwide, whatever the local name), and taxonomy (the science that does both). They discuss importance of taxonomy: identifying disease vectors (<i>Anopheles</i> mosquito and malaria around Kisumu and Lake Victoria), crop pests and beneficial organisms, economic importance of Monera (nitrogen-fixing bacteria in bean roots, bacteria in yoghurt/mursik), Protocista (algae as food chain base in Lake Victoria; Plasmodium causing malaria), and Fungi (yeast in mandazi and brewing, mushrooms as food, moulds causing food spoilage), and conserving biodiversity such as Mau Forest, Kakamega Forest and Mt Kenya. 5. Learners compare their Lesson 1 sorting with the five-kingdom system, writing 3 sentences on	'Claim: Rhizopus is a fungus. Evidence: it has no chlorophyll and absorbs nutrients from stale bread. Reasoning: fungi are eukaryotic, non-photosynthetic and absorb food from their surroundings, so it fits Kingdom Fungi, not Plantae.' During the Gallery Walk, assign roles and signal rotations. Circulate and ask: 'Which piece of evidence is the strongest, and why?' WAIT TIME 10-15 seconds before calling on responders. For debate, say: 'Take a position: Spirogyra is a plant or it is not. Stand on the left or right side of the room.' Ask 3 learners from each side to defend with evidence; cold-call 6 learners in total, and invite anyone to switch sides if persuaded: 'What evidence made you move?' Say: 'We respect every idea, but we change our minds only because of evidence.' Then pose: 'Why did we need one scientific name when mahindi, bhutta and corn already exist?' WAIT TIME 10-15 seconds; cold-call 3 learners. Connect to the economic and conservation discussion: 'If a mosquito in Kisumu and a mosquito in Mombasa have different local names, how can health workers share information about malaria vectors?' Emphasise the role of road safety and responsibility for any outdoor observation. Summarise on the board: Classification + Naming + Identification = Taxonomy.	and reasoning rather than opinion; Gallery Walk gives every learner exposure to multiple arguments and peer critique; structured debate makes learners experience how scientists revise ideas with evidence. Linking back to Lesson 1 explicitly resolves the cognitive conflict of the anchoring phenomenon.	two specific observable features, (c) reasoning that links features to kingdom characteristics; peers' sticky-note feedback is specific; during debate, learners cite evidence at least once and at least some revise their position citing evidence. Learners correctly explain why binomial names are universal and give at least one economic or conservation importance of taxonomy. Short quiz (3 questions): name the five kingdoms; give the correct binomial format; state one reason taxonomy helps conservation.
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	what was useful and what was missing from their original criteria.			
Driving Question Board (DQB) Creation  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Plant Classification (Flexbook) Source: CK-12  Search ARES for similar readings	1. The class returns to the Driving Question Board started in Lesson 1, with the driving question at the centre: 'How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?' 2. Learners work in groups to review each sticky-note question posted over the previous seven lessons (e.g. 'Why do different people call the same organism by different names?', 'Why is a mushroom not a plant?', 'What does binomial mean?', 'Why do we need ranks like genus and species?'). 3. Groups decide which questions are now ANSWERED, PARTLY answered or NEW, and move notes to matching columns with a one-line evidence statement referring to the lesson in which they learned it. 4. Each learner writes one new or remaining question (e.g. 'How do scientists classify viruses?', 'What happens when new species are discovered in Kenyan forests?', 'How does DNA change classification?') and posts it under 'Questions for further study'. 5. Class co-constructs the final answer on a chart: 'Living organisms can be systematically organised into hierarchical groups (five kingdoms down to species), named with a two-part scientific name, and identified using shared observable characteristics.'	Say: 'Our board has grown for eight lessons. Let us see how far we have come.' Read the central driving question aloud. Assign each group a section of the board (questions from Lessons 1-2, 3-4, 5-6, 7). Ask: 'Which of our questions can we now answer with evidence? Show me the lesson and the evidence.' WAIT TIME 10-15 seconds before each group reports. Cold-call 5 learners to move a note and justify its placement. For any unanswered question, say: 'Great question; where might we find the answer? A library, a museum like the National Museums of Kenya, a biology teacher, a later lesson?' Praise question-asking: 'Good scientists end with better questions.' Ensure learners from different backgrounds see their local-name questions acknowledged. Use the final co-constructed answer to bridge to the Model Building phase.	Driving Question Board: making the learning journey visible by sorting questions into answered / partly answered / new. This builds ownership, shows progress, and keeps the anchoring phenomenon central. Peer teaching occurs as groups explain how their evidence answers earlier questions.	Observation and oral questioning. Indicators: learners link an answer to a specific piece of evidence or lesson; at least 80% of notes are accurately categorised; every learner contributes a new or follow-up question; learners correctly use vocabulary (taxon, kingdom, genus, species, binomial nomenclature) when explaining notes. Exit ticket (collected at the end of the lesson): 'One question I can now answer is ___ because ____.' One question I still have is ____.'
Model Building Phase  VIDEO: 24 Six Kingdoms	1. Learners take out their Lesson 1 initial model/poster (their original classification of goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito)	Say: 'Take out your Lesson 1 model. In your own words, what did you believe then? Now build the model that explains the whole puzzle.' Provide chart paper,	Model revision + Claim-Evidence-Reasoning: comparing Lesson 1 and Final models makes conceptual change explicit and shows how evidence replaced	Final classification challenge, taxonomic hierarchy diagrams, worksheets, group presentations and peer assessment. Indicators: (1) Final Model correctly shows

Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  EXERCISE: Plant Classification Practice Source: CK-12 Search ARES for similar readings	and their prediction sheets. 2. Individually, then in groups, they build a Final Model, a concept map/poster including: the five kingdoms with key characteristics and Kenyan examples; the taxonomic hierarchy from kingdom to species; a binomial name example for a Kenyan organism (e.g. Zea mays, Homo sapiens, Anopheles gambiae, Oreochromis niloticus); the role of taxonomy in disease control, agriculture/food and biodiversity conservation. 3. Learners place the original eight organisms into the five kingdoms with evidence and write their binomial names where known. 4. Model comparison: using a two-column table 'Lesson 1 model vs Final model', learners record what changed (criteria, groups, names) and why, citing evidence. 5. Final Classification Challenge: individually, each learner receives 3 new unfamiliar organism cards, classifies them into kingdoms with CER statements and a correct binomial format. 6. Final Explanation (written, 8-10 sentences): 'How can living organisms be systematically organised, named and identified?' using the sentence starters: 'Organisms are organised by __; they are named using __; we identify them by __; this matters because __.' 7. Learners present their charts of the taxonomic hierarchy to peers, building self-efficacy, and complete a peer assessment.	markers and the evidence tables. Circulate and prompt: 'Which part of your Lesson 1 model would a scientist in Nairobi, Kisumu and Mombasa all agree on? Which part would they not?' WAIT TIME 10-15 seconds after each prompt. Cold-call 4 groups to explain one change they made and what evidence caused it. Read the Final Explanation criteria aloud: (1) mentions classification into kingdoms using shared characteristics; (2) mentions binomial naming; (3) mentions identification using observable features; (4) gives at least one importance of taxonomy (disease control, food/agriculture, conservation); (5) refers to the Lesson 1 puzzle. Support learners needing language help with sentence starters and bilingual vocabulary (Kiswahili/vernacular terms) and allow oral recording of explanations where writing is a barrier. Say: 'Your confidence comes from your evidence, so present as taxonomists.' Close by asking 3 volunteers to read the Final Explanation, then restate the driving question and answer: 'Living organisms are organised into hierarchical groups, from five kingdoms down to species, given a universal two-part scientific name, and identified through shared observable characteristics. This is the work of taxonomy.' Remind learners to apply road safety and care for the environment if they carry out any follow-up observations.	personal criteria with a universal system. The final written explanation integrates classification, naming and taxonomy into one coherent causal account answering the driving question.	five kingdoms with characteristics and examples and a correctly ordered hierarchy; (2) at least 2 of 3 new organisms correctly classified with evidence and correct binomial format; (3) model comparison identifies at least two specific changes with reasons; (4) Final Explanation meets at least 4 of 5 criteria; (5) presentation is clear and confident, and peer assessment forms are completed with respectful, evidence-based feedback. Learners scoring below 3 criteria receive follow-up support using the card sort.
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D. TEACHER REFLECTION

Which learners moved from classifying by habitat, colour or usefulness to using structural evidence such as cell type, cell wall and mode of nutrition? Did the Jigsaw ensure every learner, including quieter ones, contributed knowledge? Which misconceptions persisted (mushroom as plant, algae as plant or animal, bacteria as all harmful, binomial names written incorrectly), and what evidence helped to resolve them? Did the use of local names (Kikuyu, Luo, Kiswahili) alongside scientific names make the need for a universal system clearer? How well did learners connect taxonomy to disease control, food security and conservation in their own communities? Were practical constraints (printed photos, shared textbooks, prepared slides, unreliable electricity) managed so that no learner was disadvantaged? What would I change to give more time for the Final Explanation and Final Classification Challenge?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that unfamiliar organisms such as Amoeba, Spirogyra, Rhizopus, a mushroom, bacteria, maize, a tilapia and an Anopheles mosquito could be placed into one of five kingdoms (Monera, Protocista, Fungi, Plantae, Animalia) using shared observable evidence: cell type, number of cells, cell wall, chlorophyll, and mode of nutrition. We also observed that our Lesson 1 groupings by habitat, colour, size or usefulness placed the same organisms in different groups and local names differed.
What did I learn?	We learned that classification groups organisms using shared characteristics in a hierarchy (kingdom, phylum, class, order, family, genus, species); that binomial nomenclature gives every species a unique two-part scientific name (genus capitalised, species lowercase) understood worldwide; and that taxonomy is important for identifying disease vectors, crop pests and useful organisms, understanding the economic importance of Monera, Protocista and Fungi, and conserving biodiversity.
How does this explain the phenomenon?	Living organisms can be systematically organised into hierarchical groups from five kingdoms down to species, named using a universal two-part scientific name, and identified through shared observable characteristics. Because everyone uses the same evidence-based criteria and names, people in Nairobi, Kisumu, Mombasa and across the world can identify, study and communicate about the same organism in the same way, which resolves the puzzle from Lesson 1.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.